Pimpri Chinchwad Education Trust's

PIMPRI CHINCHWAD COLLEGE OF ENGINEERING

SECTOR NO. 26, PRADHIKARAN, NIGDI, PUNE 411044

(An Autonomous Institute Approved by AICTE and Affiliated to SPPU, Pune)

Curriculum Structure and Syllabus

of

Second Year B. Tech. Computer Science & Engineering (Artificial Intelligence & Machine Learning)

(Regulations 2023)

Effective from Academic Year 2024-25

Institute Vision

To be one of the top 100 Engineering Institutes of India in coming five years by offering exemplarily Ethical, Sustainable and Value Added Quality Education through a matching ecosystem for building successful careers.

Institute Mission

1. Serving the needs of the society at large through establishment of a state-of-art Engineering Institute.
2. Imparting right Attitude, Skills, Knowledge for self-sustenance through Quality Education.
3. Creating globally competent and Sensible engineers, researchers and entrepreneurs with an ability to think and act independently in demanding situations.

EOMS Policy

"We at PCCOE are committed to offer exemplarily Ethical, Sustainable and Value Added Quality Education to satisfy the applicable requirements, needs and expectations of the Students and Stakeholders.

We shall strive for technical development of students by creating globally competent and sensible engineers, researchers and entrepreneurs through Quality Education.

We are committed for Institute's social responsibilities and managing Intellectual property.

We shall achieve this by establishing and strengthening state-of-the-art Engineering Institute through continual improvement in effective implementation of Educational Organizations Management Systems (EOMS)."

Course Approval Summary

Board of Studies - Department of Computer Science & Engineering(AI&ML)

Sr. No.	Name of the Course	Course Code	Page Number	Signature and stamp of BoS Chairman
1.	Machine Learning	BCS23PC02	11	
2.	Data Structures & Algorithms	BCS23PC03	13	
3.	Object Oriented Programming Laboratory	BCS23PC04	45	
4.	Data Structures Laboratory	BCS23PC05	48	
5.	Community Engineering Project	BCS23EL01	54	
6.	Theory of Computation	BCS24PC06	59	
7.	Database Management Systems	BCS24PC07	61	
8.	Database Management Systems Laboratory	BCS24PC08	83	
9.	Advanced Data Structures Laboratory	BCS24PC09	85	
10.	Full Stack Development -I	BCS24VS03	88	

Chairman, Board of Studies
Computer Science & Engineering (AI & ML)
PCET's Pimpri Chinchwad College of Engineering
Sector 26, Pradhikaran, Nigdi, Pune - 411044.

Board of Studies – Applied Science and Humanities

Sr. No.	Name of the Course	Course Code	Page Number	Signature and stamp of BoS Chairman
1.	Business Studies for Engineers	BSH23EM01	43	
2	Universal Human Values	BSH23VE01	51	
3.	Statistical Data Analysis using R	BSH23OE04	15	
4.	Advanced Materials and Characterizations	BSH23OE05	17	
5.	Neural Network and Fuzzy Logic Control	BSH23OE06	31	
6.	Applied Mathematics (Suggested for Mechanical)	BSH23OE01	63	
7.	Computational Techniques (Suggested for E&TC)	BSH23OE02	65	
8.	Applied Mathematics (Suggested for Civil)	BSH23OE03	67	
9.	Computational Techniques (Suggested for Computer and IT)	BSH24OE07	69	
10.	Mathematical Optimization (Suggested for CSE -AIML)	BSH24OE08	71	
11.	Designing Thinking & Innovation Management	BSH24EM02	73	
12.	Project Management	BSH24EM03	75	
13.	Fostering Entrepreneurship and Startups.	BSH24EM04	77	
14.	Business Finance for Engineers	BSH24EM05	79	
15.	Professional Development Training	BSH24AE05	81	
16.	Constitution of India	BSH24VE02	90	

Chairman
BoS, Applied Sciences & Humanities
PCET's, Pimpri Chinchwad College of Engineering
Sector No. 26, Pradhikaran, Nigdi, Pune-44

Board of Studies – Applied Science and Humanities

Sr. No.	Name of the Course	Course Code	Page Number	Signature and stamp of BoS Chairman
1.	Business Studies for Engineers	BSH23EM01	43	
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6.	Applied Mathematics (Suggested for Mechanical)	BSH23OE01	63	
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8.	Applied Mathematics (Suggested for Civil)	BSH23OE03	67	
9.	Computational Techniques (Suggested for Computer and IT)	BSH24OE07	69	
10.	Mathematical Optimization (Suggested for CSE -AIML)	BSH24OE08	71	
11.	Designing Thinking & Innovation Management	BSH24EM02	73	
12.	Project Management	BSH24EM03	75	
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15.	Professional Development Training	BSH24AE05	81	
16.	Constitution of India	BSH24VE02	90	

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BoS, Applied Sciences & Humanities
PCET's, Pimpri Chinchwad College of Engineering
Sector No. 26, Pradhikaran, Nigdi, Pune-44

Board of Studies – Civil Engineering

Sr. No.	Name of the Course	Course Code	Page Number	Signature and stamp of BoS Chairman
1.	E-Waste Management	BCI23OE01	19	Chairman BoS, Civil Engineering PCET's, Pimpri Chinchwad College of Engineering Sector No. 26, Pradhikaran, Nigdi, Pune-44
2.	Total Quality Management	BCI23OE02	33	
3.	Building Services and Maintenance	BCI23OE03	35	

Board of Studies – Electronics & Telecommunication Engineering

Sr. No.	Name of the Course	Course Code	Page Number	Signature and stamp of BoS Chairman
1.	Electrical Machines	BET23OE01	21	Chairman BoS, Electronics & Telecommunication Engineering PCET's, Pimpri Chinchwad College of Engineering Sector No. 26, Pradhikaran, Nigdi, Pune-44
2.	Introduction to Signals and Systems	BET23OE02	23	
3.	Biology for Engineers	BET23OE03	37	

Board of Studies – Mechanical Engineering

Sr. No.	Name of the Course	Course Code	Page Number	Signature and stamp of BoS Chairman
1.	Material Science	BME23OE01	25	Chairman BoS, Mechanical Engineering PCET's, Pimpri Chinchwad College of Engineering Sector No. 26, Pradhikaran, Nigdi, Pune-44
2.	Drawing for Engineers	BME23OE02	27	
3.	Fundamentals of Mechanical Components and Systems	BME23OE03	29	
4.	Industry 4.0	BME23OE04	39	
5.	Energy storage management	BME23OE05	41	

Approved by Academic Council:

"Knowledge Brings Liberation"

Chairman
Academic Council
PCET's, Pimpri Chinchwad College of Engineering
Sector No. 26, Pradhikaran, Nigdi, Pune-44

Chairman, Academic Council
Pimpri Chinchwad College of Engineering

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CURRICULUM FRAMEWORK
(Regulations 2023)

LIST OF ABBREVIATIONS:

Sr. No.	Abbreviation	Type of Course
1	BSC	Basic Science Course
2	ESC	Engineering Science Course
3	PCC	Programme Core Course
4	PEC	Programme Elective Course
5	MDM	Multidisciplinary Minor
6	OEC	Open Elective
7	VSEC	Vocational and Skill Enhancement Course
8	AEC	Ability Enhancement Course
9	EEM	Entrepreneurship/Economics/Management Course
10	IKS	Indian Knowledge System
11	VEC	Value Education Course
12	ELC	Experiential Learning Courses
13	LLC	Liberal Learning Courses

COURSE WISE CREDIT DISTRIBUTION:

Sr. No.	Type of Course	No. of Courses	Total Credits	
			NO.	%
1.	Basic Science Course	8	14	8.75
2.	Engineering Science Course	5	12	7.5
3.	Programme Core Course	21	44	27.5
4.	Programme Elective Course	10	20	12.5
5.	Multidisciplinary Minor	6	14	8.75
6.	Open Elective	4	8	5
7.	Vocational and Skill Enhancement Course	4	8	5
8.	Ability Enhancement Course	2	4	2.5
9.	Entrepreneurship/ Economics / Management Course	2	4	2.5
10.	Indian Knowledge System	1	2	1.25
11.	Value Education Course	2	4	2.5
12.	Experiential Learning Courses	4	22	13.75
13.	Liberal Learning Courses	2	4	2.5
	Total	**71**	**160**	**100**

SEMESTER-WISE COURSE DISTRIBUTION:

Course Distribution: Semester Wise										
Sr. No.	Type of Course	No. of Courses / Semester								Total
		1	2	3	4	5	6	7	8	
1.	Basic Science Course	4	4	-	-	-	-	-	-	8
2.	Engineering Science Course	3	2	-	-	-	-	-	-	5
3.	Programme Core Course	-	1	4	4	4	4	4	-	21
4.	Programme Elective Course	-	-	-	-	2	4	4	-	10
5.	Multidisciplinary Minor	-	-	1	1	2	1	1	-	6
6.	Open Elective	-	-	2	1	1	-	-	-	4
7.	Vocational and Skill Enhancement Course	1	1	-	1	-	1	-	-	4
8.	Ability Enhancement Course	1	-	-	1	-	-	-	-	2
9.	Entrepreneurship / Economics / Management Course	-	-	1	1	-	-	-	-	2
10.	Indian Knowledge System	-	1	-	-	-	-	-	-	1
11.	Value Education Course	-	-	1	1	-	-	-	-	2
12.	Experiential Learning Courses	-	-	1	-	-	-	-	3	4
13.	Liberal Learning Courses	1	1	-	-	-	-	-	-	2
Total		**10**	**10**	**10**	**10**	**9**	**10**	**9**	**3**	**71**

SEMESTER-WISE CREDIT DISTRIBUTION

Credit Distribution: Semester Wise										
Sr. No.	**Type of Course**	**No. of Credits / Semester**								**Total**
		1	**2**	**3**	**4**	**5**	**6**	**7**	**8**	
1.	Basic Science Course	7	7	-	-	-	-	-	-	14
2.	Engineering Science Course	7	5	-	-	-	-	-	-	12
3.	Programme Core Course	-	2	8	8	10	8	8	-	44
4.	Programme Elective Course	-	-	-	-	4	8	8	-	20
5.	Multidisciplinary Minor	-	-	2	2	4	2	4	-	14
6.	Open Elective	-	-	4	2	2	-	-	-	8
7.	Vocational and Skill Enhancement Course	2	2	-	2	-	2	-	-	8
8.	Ability Enhancement Course	2		-	2	-	-	-	-	4
9.	Entrepreneurship/Economics / Management Course	-	-	2	2	-	-	-	-	4
10.	Indian Knowledge System	-	2	-	-	-	-	-	-	2
11.	Value Education Course	-	-	2	2	-	-	-	-	4
12.	Experiential Learning Courses	-	2	-		1				22
13.	Liberal Learning Courses	2	2	-	-	-	-	-	-	4
Total		**20**	**20**	**20**	**20**	**20**	**20**	**20**	**20**	**160**

Curriculum Structure

Second Year B.Tech.

Computer Science & Engineering

(Artificial Intelligence & Machine Learning)

CURRICULUM STRUCTURE

Second Year B.Tech. (Computer Science & Engineering - AI&ML) Semester –III

Second Year B.Tech Computer Science & Engineering(AI&ML)- (Regulations 2023) (With effect from Academic Year 2024-25)																
Semester III																
Course Code	Course Name	Credit Scheme				Teaching Scheme (Hours / Week)			Evaluation Scheme and Marks							
		L	P	T	Total	L	P	T	FA		SA	TW	PR	OR	Total
									FA1	FA2					
BCS23PC02	Machine Learning	2	-	-	2	2	-	-	10	10	30	-	-	-	50
BCS23PC03	Data Structures & Algorithms	3	-	-	3	3	-	-	20	20	60	-	-	-	100
-	MDM: Course 1*	2	-	-	2	2	-	-	10	10	30	-	-	-	50
-	Open Elective-Engineering Science	2	-	-	2	2	-	-	10	10	30	-	-	-	50
-	Open Elective-Department Specific Electives	2	-	-	2	2	-	-	10	10	30	-	-	-	50
BSH23EM01	Business Studies for Engineers	2	-	-	2	2	-	-	10	10	30	-	-	-	50
BCS23PC04	Object Oriented Programming Laboratory	-	1	-	1	-	2	-	-	-	-	25	-	25	50
BCS23PC05	Data Structures Laboratory	-	2	-	2	-	4	-	-	-	-	50	50	-	100
BSH23VE01	Universal Human Values	2	-	-	2	2	-	-	20	30	-	-	-	-	50
BCS23EL01	Community Engineering Project	-	2	-	2	-	4	-	-	-	-	50	-	50	100
	Total	15	5	-	20	15	10	-	-	-	-	-	-	-	650

L-Lecture, **P**-Practical, **T**-Tutorial, **FA**– Formative Assessment, SA- Summative Assessment, **TW**-Term Work, **OR**-Oral, **PR**-Practical

*** For MDM Course 1 syllabus refer to the respective MDM Syllabus booklet.**

List of courses - Open Elective (Engineering Science)

Course Code	Course Name	
BSH23OE04	Statistical Data Analysis using R	Choose any one
BSH23OE05	Advanced Materials and Characterizations	
BCI23OE01	E-Waste Management	
BET23OE01	Electrical Machines	
BET23OE02	Introduction to Signals and Systems	
BME23OE01	Material Science	
BME23OE02	Drawing for Engineers	
BME23OE03	Fundamentals of Mechanical Components and Systems	

List of courses - Open Elective (Department Specific Electives)

Course Code	Course Name	
BSH23OE06	Neural Network and Fuzzy Logic Control	Choose any one
BCI23OE01	Total Quality Management	
BCI23OE02	Building Services and Maintenance	
BET23OE03	Biology for Engineers	
BME23OE04	Industry 4.0	
BME23OE05	Energy storage management	

Second Year B.Tech. (Computer Science & Engineering - AI&ML) Semester – IV

Second Year B.Tech Computer Science & Engineering(AI&ML)- (Regulations 2023) (With effect from Academic Year 2024-25)															
Semester IV															
Course Code	Course Name	Credit Scheme				Teaching Scheme (Hours / Week)			Evaluation Scheme and Marks						
		L	P	T	Total	L	P	T	FA		SA	TW	PR	OR	Total
									FA1	FA2					
BCS24PC06	Theory of Computation	2	-	1	3	2	-	1	20	20	60	-	-	-	100
BCS24PC07	Database Management Systems	2	-	-	2	2	-	-	10	10	30	-	-	-	50
-	MDM: Course 2*	2	-	-	2	2	-	-	10	10	30	-	-	-	50
-	Open Elective-Mathematics	2	-	-	2	2	-	-	10	10	30	-	-	-	50
-	EEM-II	2	-	-	2	2	-	-	10	10	30	-	-	-	50
BSH24AE05	Professional Development Training	-	2	-	2	-	4	-	-	-	-	100	-	-	100
BCS24PC08	Database Management Systems Laboratory	-	2	-	2	-	4	-	-	-	-	50	50	-	100
BCS24PC09	Advanced Data Structures Laboratory	-	1	-	1	-	2	-	-	-	-	25	25	-	50
BCS24VS03	Full Stack Development -I	-	2	-	2	-	4	-	-	-	-	50	-	50	100
BSH24VE02	Constitution of India	2	-	-	2	2	-	-	20	30	-	-	-	-	50
	Total	12	7	1	20	12	14	1							700

L-Lecture, **P**-Practical, **T**-Tutorial, **CIE**–Continuous Internal Evaluation, **IE**-Internal Evaluation, **ETE**-End Term Evaluation, **TW**-Term Work, **OR**-Oral, **PR**-Practical

*** For MDM Course 2 syllabus refer to the respective MDM Syllabus booklet.**

Note:- Exit Policy: Available as a separate document.

List of courses - Open Elective (Mathematics)

Course Name	Course Code	
BSH24OE01	Applied Mathematics (Suggested for Mechanical branch)	Choose any one
BSH24OE02	Computational Techniques (Suggested for E&TC)	
BSH24OE03	Applied Mathematics (Suggested for Civil)	
BSH24OE07	Computational Techniques (Suggested for Computer and IT)	
BSH24OE08	Mathematical Optimization (Suggested for CSE -AIML)	

List of courses - EEM-II (Entrepreneurship , Economics, Management Courses)

Course Code	Course Name	
BSH24EM02	Designing Thinking & Innovation Management	Choose any one
BSH24EM03	Project Management	
BSH24EM04	Fostering Entrepreneurship and Startups.	
BSH24EM05	Business Finance for Engineers	

Course Syllabus

Second Year B.Tech.

Computer Science & Engineering

(Artificial Intelligence & Machine Learning)

Semester - III

Program:	CSE (AI & ML)					Semester:	III
Course :	Machine Learning					Code :	BCS23PC02
Credits	Teaching Scheme (Hrs./Week)			Evaluation Scheme and Marks			
	Lecture	Practical	Tutorial	FA		SA	Total
				FA1	FA2		
2	2	-	-	10	10	30	50

Prior knowledge of Data Science is essential.

Course Objectives:

This course aims at enabling students,

1. To understand the basics of Machine Learning.
2. To apply different machine learning models to various datasets.
3. To design and implement various machine learning algorithms in a range of real-world applications.
4. To interpret the model results and analyze the accuracy of the model.

Course Outcomes:

After learning the course, the students should be able to:

1. Use different pre-processing techniques and solve regression problems.
2. Analyze various supervised Machine Learning algorithms and Evaluation metrics.
3. Apply unsupervised ML algorithms to real time datasets & Evaluate the performance
4. Apply reinforced and Ensemble learning algorithms for different ML problems.

Detailed Syllabus:

Unit	Description	Duration [Hrs]
I	**Basics of Machine Learning** Introduction to Machine Learning, Applications, Supervised, Unsupervised and Reinforced learning, Feature Selection, Feature reduction using PCA, Linear Regression, Ridge regression, Lasso and ElasticNet Regression, Polynomial regression, Gradient Descent Algorithm	7
II	**Supervised Learning** Supervised Classification Algorithms: K-Nearest Neighbor, Decision Tree classification, Random Forest, Support Vector Machine, Evaluation Measures: SSE, MME, R2, Confusion Matrix, Precision, Recall, F-score, ROC-Curve, Cross-validation	8
III	**Unsupervised Clustering** Introduction to clustering, Applications, Types of Clustering: Hierarchical, Agglomerative clustering and Divisive clustering, Partitional Clustering- K-means clustering, Elbow Method, Associative rule Learning	7

IV	**Reinforcement Learning, Ensemble Learning & ANN** Reinforcement Learning through feedback network, Function Approximation, Ensemble Method: Bagging, boosting, stacking and learning with ensemble, Introduction to ANN: Perceptron model, MLP, Backpropagation Algorithm.	**8**
	Total	**30**

Text Books:

1. Bishop M., C.," Pattern Recognition and Machine Learning", Springer-Verlag (2011) 2nd Edition.
2. Mitchell M., T., "Machine Learning", McGraw Hill (1997) 1st Edition, ISBN : 0070428077978007042807 2
3. Alpaydin E.," Introduction to Machine Learning", MIT Press (2014) 3rd Edition.

Reference Books:

1. Michie D., Spiegelhalter J. D., Taylor C. C., Campbell, J., "Machine Learning, Neural and Statistical Classification", Overseas Press (1994).
2. Duda, Hart, Stork, Pattern Classification, Wiley, 2001

e-sources:

NPTEL Video Courses:

1. Introduction to Machine Learning by Prof. Sudeshana Sarkar (IITKGP)
2. Introduction to Machine Learning by Prof. B. Ravindran (IITM)
3. Machine Learning and Deep Learning for Engineers and Scientists By Prof Balaji (IITM)
4. Machine Learning by Carl Gustaf Jansson, KTH The Royal Institute of Technology, Stockholm, Sweden

Program :	S.Y. B.Tech. C.S.E. (AI & ML)					Semester:	III
Course :	Data Structures & Algorithms					Code :	BCS23PC03
Credits	Teaching Scheme (Hrs./Week)			Evaluation Scheme and Marks			
	Lecture	Practical	Tutorial	FA		SA	Total
				FA1	FA2		
3	3	-	-	20	20	60	100

Prior Knowledge of : Computer Programming and Problem Solving, python is essential.

Course Objectives:

1. To understand the standard and abstract data representation methods.
2. To perform operations on the various structured data.
3. To use appropriate data structure for logic building in computational solutions.
4. To understand various data searching and sorting methods with pros and cons.
5. To develop a logic for graphical modeling of real life problems.

Course Outcomes :

After learning the course, students will be able to:

1. Analyze algorithms through frequency count, time complexity, and space complexity metrics.
2. Apply concepts of linked lists to solve real world problems.
3. Solve problems related to data organization and manipulation using stacks and queues effectively.
4. Understand and apply concepts of binary trees for solving problems.
5. Analyze and solve problems related to graph structures and their applications.
6. Analyze, select and apply appropriate search, sorting, and hashing techniques based on problem requirements.

Detailed Syllabus

Unit	Description	Duration [Hrs]
I	**Introduction to Data structures and Array:** Types of Data Structure - Linear & Nonlinear, Static & Dynamic **Algorithms:** Characteristics of algorithms, Analysis of algorithms – Frequency Count, Time & Space complexity, Array: Memory representation, Array as an abstract data type (ADT), Types One Dimensional Array & 2 Dimensional Array, Address Calculation, Operations on Array.Applications: Polynomial representation using array & array of structure.	7
II	**Linked List:** Dynamic Memory Management, Basics of Linked List, Comparison of sequential and linked organizations, Types of linked list, Singly linked list, Doubly linked list, Circular linked list. Operations on Linked list-insertion, deletion	8
III	**Stack & Queue:** Fundamentals of stack, Stack representation using array and linked List, Operations on stack.**Applications**: Recursion, Validity of parentheses, Expression Evaluation and Conversion. Fundamentals of queue, Queue representation using array and Linked List, Types of queue – Circular Queue, Dequeue, Priority Queue	7

IV	**Tree:** Basic terminology, representation using array and linked list, Recursive And Non recursive Tree Traversals, Insertion, deletion operations on binary tree, Construction of binary tree from traversals, Binary Search tree (BST): Insertion, deletion of a node from BST. Threaded Binary tree (TBT): Creation and traversals on TBT. Height Balanced Tree- AVL tree.	**8**
V	**Graph:** Basic Concepts, Storage representation, Adjacency matrix, adjacency list, adjacency multi list, inverse adjacency list. Traversals-depth first and breadth first search, Applications : Minimum spanning Tree using Prim's and Kruskal's Algorithm	**8**
VI	**Searching , Sorting Techniques and Hashing:** Search Techniques - Linear search, Binary search. Sorting - Insertion sort, Selection sort. Quick sort and Merge Sort. Hashing: Concepts - Hash table, issues in hashing, hash functions- properties of good hash function, Collision resolution strategies- open addressing and chaining	**7**
Total		45

Text Books:

1. Ellis Horowitz, Sartaj Sahni, Dinesh Mehta, Fundamentals of Data structures in C++, 2nd Edition, University Press (India) Pvt.Ltd. 2008
2. Horowitz and Sahani, "Fundamentals of Data Structures in C++", University Press, ISBN 10:0716782928 /ISBN 13: 9780716782926.
3. Brassard & Bratley, "Fundamentals of Algorithms", Prentice Hall India/Pearson Education, ISBN 13-9788120311312.

Reference Books:

1. G.A.V. PAI, Data Structures and Algorithms, Concepts, Techniques and Applications, Volume1,1st Edition , Tata McGraw-Hill,2008.
2. Richard F. Gilberg& Behrouz A. Forouzan, Data Structures, Pseudocode Approach with C, 2nd Edition, Cengage Learning India Edition,2007.

e-sources:

1. https://www.cs.usfca.edu/~galles/visualization/Algorithms.html

Program :	B. Tech. CSE (AI & ML)					Semester: III	
Course :	OE-ES- Statistical Data Analysis Using R (Offered by AS&H)					Code :	BSH23OE04
Credits	Teaching Scheme (Hrs./Week)			Evaluation Scheme and Marks			
	Lecture	Practical	Tutorial	FA		SA	Total
				FA1	FA2		
2	2	-	-	10	10	30	50

Prior knowledge of: Basics of Statistics and Probability **is essential.**

Course Objectives:

After completion of the course, students will have an adequate background, conceptual clarity, and knowledge of mathematical principles related to data, preprocessing techniques for data visualization, and statistical techniques for prediction and decision-making.

Course Outcomes:

After learning the course, the students should be able to:

1. **Understand** the data and differentR packages related to data science and its access.
2. **Make use of** data pre-processing methods and generate quality data for analysis.
3. **Apply** different data visualization techniques to understand the data.
4. **Analyse** the data for decision-making using statistical methods.

Detailed Syllabus:

Unit	Description	Duration [Hrs]
I	**Fundamentals of R Software for Data** Introduction to Data: Definition, Types and Properties, R Packages for Data Science, Importing and Exporting Data in R Software, Accessing Databases with R Software.	7
II	**Data Wrangling** Pre-processing Data in R Software, Dealing with Missing Values in R Software, Data Formatting in R Software, Data Normalization in R Software, Binning in R Software, and Conversion of type of data	7
III	**Data Visualization** Data visualisation for various data categories like Categorical, Numerical, or both using plots likeHistogram, Bar/ Line Chart, Box Plots (including group-by option), Scatter Plots, Mosaic Plot, etc., and their interpretations	8
IV	**Data Analysis** Sampling, Descriptive Statistics, Linear regression and Multiple linear regression. Model evaluation using visualization, prediction, and decision-making.	8
	Total	**30**

Reference Books:

1. Montgomery and Runger, “Applied Statistics and Probability for Engineers”, Wiley, India, 6 Edition, ISBN: 9788126562947.
2. R. Johnson, “Probability and Statistics for Engineers”, Prentice India Ltd, 8 Edition, ISBN 13:978-8120342132.
3. S.P. Gupta, “Statistical Methods”, Paperback publication, 43 edition, ISBN: 9788180549892, 8180549895.
4. Victor A. Bloomfield, “Using R for Numerical Analysis in Science and Engineering”, CRC Press, First Edition, ISBN: 9781315360492

e-sources:

NPTEL Course lectures links:

1. https://www.youtube.com/watch?v=VVYLpmKRfQ8&list=PL6C92B335BD4238AB (Probability)
2. https://nptel.ac.in/courses/111104100 (Introduction to R software)
3. https://www.youtube.com/watch?v=WbKiJe5OkUU&list=PLFW6lRTa1g83jjpIOte7RuEYCwOJa-6Gz (Descriptive statistics using R software)

Program :	**B. Tech. CSE (AI & ML)**					**Semester:**	**III**
Course :	**OE-ES- Advanced Materials and Characterizations** (offered by AS&H)					**Code :**	**BSH23OE05**
Credits	**Teaching Scheme (Hrs./Week)**			**Evaluation Scheme and Marks**			
	Lecture	**Practical**	**Tutorial**	**FA**		**SA**	**Total**
				FA1	**FA2**		
02	**02**	**-**	**-**	**10**	**10**	**30**	**50**

Prior knowledge of basic physics, chemistry and nanotechnology is essential**.**

Course Objectives: This course aims at enabling students,
1. To learn the principles of advanced materials, technologies and characterizations.
2. To undertake research projects with applications of advanced materials.

Course Outcomes:
After learning the course, the students should be able to:
1. Interpret structure, properties and applications of advance engineering materials.
2. Explain the properties and requirements of materials for some advanced applications
3. Analyze structural, optical, elemental & morphological properties of the materials
4. Interpret electrochemical & thermal properties of the materials

Detailed Syllabus:

Unit	**Description**	**Duration [Hrs]**
I	**Engineering Materials** Polymer Composite eg. fiber- reinforced polymer (FRP) composites Advanced carbon materials eg. Graphene & CNT Alloys: a] Nano alloy eg. Cu-Ni nano alloy b] Memory alloy-Nitinol CdSe Thin films for solar cells.	**7**
II	**Materials for Special applications** Battery and Supercapacitor: working principles, components, Superconductors, material requirement and properties of electrodes, electrolytes and spacers, metallic, nonmetallic and ceramic superconductors, Applications, Gas Sensing: working principle and required material properties. Substrate Materials for quantum computer, Invar and Elinvar Materials: properties and applications, 2D materials: properties and their applications	**7**
III	**Structural, Optical, Elemental & Morphological Characterizations** X-Ray Diffraction and phase identification, indexing lattice parameter determination, Grain size analysis, EDAX, Electron microscopy, scanning electron microscopy, Transmission electron microscopy (TEM), Fourier Transform Infrared Spectroscopy, Band gap measurements-UV Vis- IR Spectroscopy	**8**

IV	**Electrochemical Characterization** Cyclic voltammetry: Instrumentation, current-potential relation applicable for Linear Sweep Voltammetry (LSV) and Cyclic Voltammetry (CV), interpretation of cyclic voltammograms, charging-discharging behaviors of supercapacitor and batteries. **Thermal Analysis techniques:** Thermo-gravimetric analysis (TGA),Differential thermal analysis (DTA) analysis, Thermal expansion measurements, Thermal conductivity measurements, Ionic conductivity measurements. Specific heat capacity measurements, Debye temperature measurements	**8**
	Total	**30**

Reference Books:

1. Elements of X-ray Diffraction, B.D. Culity and S.R. Stock, Pearson Publication, Third edition 2014.
2. Introduction to Fuel Cells, Electrochemistry and Materials, San Ping Jiang, Qingfeng Li, SpringerPublication, 2022.
3. Solid State Physics, S.O.Pilli, New age, International Publication Tenth edition 2022.
4. Introduction to Solid State Physics, C. Kittle, 8th edition Wiley, 2005.
5. Introduction to Superconductivity, Michael Tinkham, 2nd edition, Dover Publication 2004.
6. Electrochemical super capacitors, B. E. Conway, Springer, 1999.
7. Spectroscopy, G.R. Chatwal and S.K. Ananad, Himalaya Publications, 2016.
8. Introduction to Thermal Analysis, M.E. Brown, Kluwer academic Publishers, 2nd edition 2001.
9. Electrochemical Methods: Fundamentals and Applications, A, J Bard, Allen J. Bard, Larry R. Faulkner, Henry S. White, John Wiley & Sons, 31 May 2022.
10. Microscopy A Very Short Introduction by Srivastava, Oxford University Press, 2015.
11. Practical Guide to materials Characterization, Khalid Sultan, Wiley-VCH, 2023
12. Engineering Chemistry by Wiley India Pvt.Ltd,First edition 2011.
13. Introduction to Nanotechnology by Charles P. Poole, Frank Owens, John Wiley & Sons (2003)

e-sources:

1. https://archive.nptel.ac.in/courses/113/106/113106034/

Program :	B. Tech. CSE (AI & ML)					Semester:	III
Course :	OE-ES- E-waste Management (Offered by Civil Department)					Code :	BCI23OE01
Credits	Teaching Scheme (Hrs./Week)			Evaluation Scheme and Marks			
	Lecture	Practical	Tutorial	FA		SA	Total
				FA1	FA2		
2	2	-	-	10	10	30	50

Prior Knowledge:

1. Fundamentals of environmental science.
2. Fundamentals of sustainable development.

Course Objectives: This course aims at enabling students,

1. To impart the knowledge of issues and challenges of e-waste management.
2. To create awareness of potential health effects and risk associated with e-waste.
3. To build knowledge of e-waste legislation (policy and guidelines) and circular economy.
4. To get acquainted with recycling, recovering, and disposal techniques.

Course Outcomes:

After learning the course, the students should be able to:

1. Identify the issues and challenges of e-waste management for a sustainable environment.
2. Analyze potential health effects and risk assessment.
3. Illustrate e-waste laws and guidelines and apply a circular economy road map for an e-waste sustainable future.
4. Identify the e-waste recycling, recovery and disposal techniques and its significance for a sustainable future.

Detailed Syllabus:

Unit	Description	Duration [Hrs]
I	**Electronic Waste Management – Issues and Challenges:** Introduction to e-waste, classification and composition, need to manage / recycle, E-waste generation in India and comparison with world scenario; facts & figures, estimation of waste electronic and electrical equipment (WEEE), economic assessment of E-waste (Rare earth minerals, precious metals), effluents (solid, liquid and gas) generated during recycling, quantification of health hazard due to informal recycling of E-Waste,(Case study-based learning)	7
II	**E-Waste: Environmental and Public health issue:** Characteristics of Hazardous Substances, toxicity concerns, potential health effects and symptoms of long-term exposure, case study of heavy metal contamination due to E-Waste recycling, Introduction to risk assessment, epidemiologic data analysis and parameter for determining exposure and disease (numerical), classification of potential carcinogens, dose-response assessment, Potency Factor for Carcinogens (numerical), Hazard Index (HI) and Hazard Quotient (HQ) (numerical).	8

III	**E-waste (Management & Handling) Rules / Guidelines and Circular Economy:** Regulatory frameworks in India, objectives of e-waste rules, hazardous and other wastes (Management and Transboundary Movement) Rules, 2016, application of rules to stakeholders, objectives of e-waste rules, India's stand on liberalizing import rules, UN Sustainable Development Goals (SDGs) and e-Waste, circular economy startup in India with a case study.	**7**
IV	**Recycling and Recovery of Metals from Electronic Waste and disposal techniques:** E-waste recycling machineries, recycling process of E-Waste, existing E-Waste recycling Techniques, case study/ examples (metal recovery process), mechanism of extraction of precious metal from leaching solution, recovery of precious metals from solutions by solvent extraction, extraction of precious and rare earth metals from End-of- Life (EOL) electronic products, disposal techniques, role and responsibility of extended producers' responsibility (EPR), E-waste economy in the organized and unorganized sector, Case study on recycling and precious metal recovery from e-waste.	**8**
	Total	**30**

Text Books:

1. E-waste management challenges and opportunities in India, Varsha Bhagat Ganguly, Routledge India, 1st edition, 2021.
2. E-waste Management and procurement of Environment, Dr. Suresh Kumar and Dr. Jitendra Kumar Pradhan, Authorpress, 2021 edition.
3. E-waste in India: Management, challenges and opportunities (Volume I & II), Dr. Suresh Kumar, Authorspress, September, 2021 edition.

Reference Books:

1. Fowler B, Electronic Waste – 1st Edition (Toxicology and Public Health Issues), 2017 Elsevier
2. Johri R., E-waste: implications, regulations, and management in India and current global best practices, The Energy and Resources Institute (TERI)TERI Press, New Delhi, 2008.
3. The Complete Technology Book on E-Waste Recycling (Printed Circuit Board, LCD, Cell Phone, Battery, Computers), ASIA PACIFIC BUSINESS PRESS Inc., 2015.

e-sources:

1. https://cpcb.nic.in/e-waste/
2. https://courses.iid.org.in/course/e-waste-recycling-business
3. https://www.suritex.co.in/
4. http://greenscape-eco.com/
5. https://onlinecourses.nptel.ac.in/noc20_ce12/preview
6. https://nielit.gov.in/gangtok/content/paid-course-e-waste-management
7. https://www.semanticscholar.org/paper/Electronic-waste-management-Hester Harrison/bc34471b0f9d94b0656b43df6b322116f2a7175c

Program :	B. Tech. CSE (AI & ML)					Semester:	III
Course :	OE -ES- Electrical Machines (offered by E&TC)					Code :	BET23OE01
Credits	Teaching Scheme (Hrs./Week)			Evaluation Scheme and Marks			
	Lecture	Practical	Tutorial	FA		SA	Total
				FA1	FA2		
2	2	-	-	10	10	30	50

Prior knowledge of Fundamental knowledge of electromagnetism & electrical parameters **is essential.**

Course Objectives:

This course aims at enabling students,

1. To impart basic knowledge for conceptual understanding of DC machines.
2. To explore the construction and performance characteristics of three phase AC machines.
3. To explore the construction and performance characteristics of single-phase AC machines.
4. To relate the applications of electrical machines to practical and industrial scenarios.

Course Outcomes:

After learning the course, the students should be able to:

1. Describe the constructional features and working principles of DC Machines.
2. Explain the constructional features and operation of three phase induction motors
3. Explain the constructional features and operation single phase induction motors
4. Relate the applications of electrical machines to their respective fields of study and industrial applications.

Detailed Syllabus:

Unit	Description	Duration [Hrs]
I	**Introduction to Electrical Machines:** Overview of electrical machines and their significance, Classification of electrical machines, Fundamental principles: Faraday's Law, Lenz's Law, and Electromagnetic Induction. **DC Machines** Working principle of DC machine as a generator and a motor; Types and constructional features; EMF equation of generator, DC motor working principle; Back EMF and its significance, torque equation; Types of D.C. motors, characteristics, Necessity of a starter for DC motor, Speed control methods of DC shunt and DC series motor and industrial applications.	8
II	**Three phase induction motor** – constructional features, working principle, Rotating magnetic field, slip ring and cage types. slip, phasor diagram, expression for mechanical power and torque, torque-slip characteristics, starting torque, full load and pull-out torque, equivalent circuit. Industrial applications.	7
III	**Single phase induction motors:** Construction of single-phase induction motor, double field revolving theory. Types of single-phase induction motors: Split phase and shaded pole type induction motors, applications. Specifications of induction motors (KW rating, rated voltage, current rating, frequency, speed, class of insulation)	8

IV	**Special Purpose Motors:** Construction, principle of working, characteristics ratings and applications of Brush less D.C. motors, Stepper motors (permanent magnet and variable reluctance type only), Permanent Magnet motor (A.C. & D.C.), SRM Switch reluctance motor.	**7**
	Total	

Text Books:

1. V. N. Mittal and Arvind Mittal,"Basic Electrical Engineering" , 2nd Edition. (McGraw-Hill),2010
2. D. P. Kothari and I. J. Nagrath, "Basic Electrical Engineering", Tata McGraw Hill, 2010

Reference Books:

1. J.B. Gupta, "Theory and Performance of Electrical Machines," S.K. Kataria & Sons.
2. A.E. Fitzgerald, Charles Kingsley, Stephen D. Umans, "Electric Machinery," McGraw-Hill Education.
3. D. C. Kulshreshtha, "Basic Electrical Engineering" , 1st Edition (Tata McGraw hill),2009
4. B. L. Theraja and A. K. Theraja S. Chand & Co. Pvt. Ltd. New Delhi, "A textbook of Electrical Technology Vol II",2020

e-sources:

1. Electrical Machines – I-https://onlinecourses.nptel.ac.in/noc20_ee60/preview
2. Electrical Machines – I-https://archive.nptel.ac.in/courses/108/105/108105017/

Program:	B. Tech. CSE (AI & ML)					Semester:	III
Course :	**OE- ES- Introductions to Signals and Systems** (Offered by E&TC)					Code :	BET23OE02
Credits	Teaching Scheme (Hrs./Week)			Evaluation Scheme and Marks			
	Lecture	Practical	Tutorial	FA		SA	Total
				FA1	FA2		
2	2	-	-	10	10	30	50

Prior knowledge of Linear Algebra & Differential Calculus **is essential.**

Course Objectives:

This course aims at enabling students,

1. To develop an understanding of students related to signal representation, classification, and operations.
2. To build the understanding of analyzing and classifying the systems and their applications
3. To apply the basic concept of Fourier transform and Laplace transform to the systems

Course Outcomes:

After learning the course, the students should be able to:

1. Represent, classify, and perform the operation on signals
2. Classify the system and utilize convolution for system analysis
3. Apply the basics of Fourier transform to analyze the signal in frequency domain
4. Apply the basics of the Laplace transform to analyze the signal in a complex frequency domain.

Detailed Syllabus:

Unit	Description	Duration [Hrs]
I	**Introduction to Signals**: Representation of Standard signals, Classification of signals: Continuous-time and discrete-time signals, Periodic Signals and non-periodic, Operations on signals: Time shifting, Time reversal, Time scaling, Amplitude scaling, Signal addition, Subtraction, Signal multiplication.	07
II	**Introduction to System**: System Definition and Application, Classification of Systems: Continuous-time and Discrete-time systems, Linear and Non-Linear systems, Time variant and Time-invariant systems, Stable and Unstable systems, Causal systems, and non-causal systems. Convolution sum using graphical method.	07
III	**Fourier Transform**: Fourier Transform (FT) representation of aperiodic continuous time (CT) signals, Evaluation of magnitude and phase response, FT of standard CT signals, Application of Fourier transform	08
IV	**Laplace Transform and Z Transform**: Definition of Laplace Transform (LT), ROC, Laplace transform of standard periodic and aperiodic functions, Inverse Laplace transform, Application of Laplace transforms.	08
	Total	30

Text Books:

1. A.V. Oppenheim, A.S. Willsky ―Signals and systems‖, Prentice-Hall signal processing series. 2nd Edition, 2015
2. A. Nagoor Kanni ―Signals and Systems‖, McGraw Hill, 2nd Edition, 2017

Reference Books:

1. B P Lathi ―Linear Systems and Signals‖, Oxford University Press, Third Edition, 2017
2. Simon Haykins and Barry Van Veen ―Signals and Systems‖, Wiley India, 2nd Edition. 2017
3. M.J. Roberts ―Signal and Systems‖, Tata McGraw Hill, Third Edition ,2019.
4. Charles Phillips ―Signals, Systems and Transforms‖, Pearson Education, 4th Edition. 2013
5. R. J. Beerends, H. G. ter Morsche ―Fourier and Laplace Transforms‖ Cambridge University Press, 2003.
6. Shaila Dinkar Apte, Signals and System, Cambridge University Press, Edition 1, 2018

Online courses Links:

1. https://onlinecourses.nptel.ac.in/noc23_ee14/preview
2. https://www.classcentral.com/course/engineering-iitbombay-signals-and-systems-part-1-2679

Program :	B. Tech. CSE (AI & ML)					Semester:	III
Course :	OE- ES- Material Science (offered by Mechanical Engineering)					Code :	BME23OE01
Credits	Teaching Scheme (Hrs./Week)			Evaluation Scheme and Marks			
	Lecture	Practical	Tutorial	FA		SA	Total
				FA1	FA2		
2	2	-	-	10	10	30	50

Prerequisite : Nil

Course Objectives:

This course aims at enabling students,

1. To Aquitaine students about materials, their properties, and structure property relationship.
2. To create awareness about the importance of materials information in engineering.

Course Outcomes:

After learning the course, the students should be able to:

1. Compare different materials based on their structures.
2. Relate the structure of materials with their properties.
3. Understand the structure and properties of metals, polymers, ceramics and composites.
4. Explore advanced materials in engineering applications.

Detailed Syllabus:

Unit	Description	Duration [Hrs]
I	Introduction to materials: Classification of materials, structure of materials :Atomic structure, crystal structure and microstructure, material properties: Physical, Mechanical, Electrical, Magnetic etc.	8
II	Metals: Ferrous, non ferrous, alloys, structure and properties of commonly used metals, applications of metals in engineering.	7
III	Polymers: Structure of polymers and their properties, Thermoplastic and thermosetting plastics, advanced polymers such as temperature resistant polymers, shape change polymers, conducting polymers Ceramics: Structure of ceramics and their thermal, electrical and magnetic properties, applications of ceramics in engineering.	8
IV	Composites: Classification of composite materials and their applications. Advanced materials-shape memory alloy, hydrogel, thermo responsive, photo responsive, magneto responsive, piezoelectric materials, Processing of advanced materials such as semiconductors.	7
	Total	**30**

Text Books:

1. Material Science and Engineering An Introduction, William D. Callister, Wiley Publication,10th Edition 2009.
2. Material Science and Metallurgy, Dr. V. D. Kodgire, Everest publishing house, 45th Edition, 2021.

Reference Books:

1. Materials for Engineering, John Martin, Woodhead Publishing Limited, CRC Press, 3 rd Edition, 2006.

Online Courses:

1. NPTEL Course : Nature and properties of materials (https://onlinecourses.nptel.ac.in/noc20_me13/course)
2. NPTEL Course : Properties of materials (https://onlinecourses.nptel.ac.in/noc20_mm13/course)

Program:	B. Tech. CSE (AI & ML)					Semester :	III
Course :	OE- ES- Drawing for Engineers (Offered by Mechanical)					Code :	BME23OE02
Credits	Teaching Scheme (Hrs./Week)			Evaluation Scheme			
	Lecture	Practical	Tutorial	FA			
				FA1	FA2	SA	Total
2	2	-	-	10	10	30	50

Prerequisite:Nil

Course Objectives:

1. Develop imagination of physical objects for communication of Engineering Drawing.
2. Develop the interpretation and manual drawing skills.
3. Develop the physical realization of the dimension of the objects.

Course Outcomes*:

After learning the course, students will be able to

1. **Understand** the basics of Engineering Graphics and **apply** the concepts in real life applications
2. **Analyze** 3D engineering objects and draw orthographic projections
3. **Analyze** 2D views of engineering objects and draw isometric views
4. **Analyze** 3D objects and draw development of lateral surfaces of solid

Detailed Syllabus

Unit	Description	Duration (H)
I	**Part A: Introduction to engineering drawing** **Importance of engineering drawing to Civil, Electrical, Electronics, IT and Computer Engineers**, introduction to drawing instrument and their uses, drawing sheet layout and its sizes, types of lines and their applications, dimensioning terminology and methods **Part B: Projection of Points, Lines and Planes** Projection: Introduction to projection and different methods of projections, Projection of Point: Introduction, Point situated in all the quadrants. Projection of lines: Introduction, Projection of lines. Projection of Planes: Introduction, Types of planes, Projection of planes. **CPU circuit diagram, Electrical/Electronic board circuit diagram, Building floor plan**	8
II	**Orthographic projections** Orthographic projection of given real life components by first angle method of projection, types of sections.	7
III	**Isometric view** Isometric axes, scale, difference between isometric projection and isometric view, isometric view of simple solids and its dimensioning.	8
IV	**Development of lateral surface of solids** Development of lateral surfaces of prism, pyramid, cylinder and cone. **3D and 2D drawing of branch specific industrial products and its lateral surface development.**	7
	Total	30

Text Books:

1. Engineering Drawing with an introduction to AutoCAD- Dhananjay A. Jolhe, Revised Edition 2017, Tata McGraw Hill publishing company Ltd. New Delhi, India
2. Engineering Drawing, Plane and solid geometry- N. D.Bhatt, 54th edition 2023, Charotor publication house.

Reference Books:

1. Engineering Drawing- M.B Shah and B.C Rana, 2nd edition 2009, Pearson Publications.
   Engineering Graphics- P.J. Shah, Revised edition 2019, S Chand Publications.
2. Fundamentals of Engineering Drawing- Warren J. Luzzader, 11th edition 2015, Prentice Hall of India New Delhi.
3. A text book of Engineering Drawing- R.K. Dhawan, Revised Edition 2019, S. Chand and company ltd. New Delhi, India

Program:	B. Tech. CSE (AI & ML)					Semester:	III
Course :	OE- ES- Fundamentals of Mechanical Components and Systems (Offered by Mechanical)					Code :	BME23OE03
Credits	Teaching Scheme (Hrs./Week)			Evaluation Scheme and Marks			
	Lecture	Practical	Tutorial	FA		SA	Total
				FA1	FA2		
2	2	-	-	10	10	30	50

Prerequisites: Nil

Course Objectives:

This course aims at enabling students,

1. Impart knowledge of mechanical engineering and describe the scope of mechanical engineering with multidisciplinary industries.
2. Impart knowledge of basic concepts of thermodynamics and heat transfer applied to industrial applications.
3. Understand, identify and get exposure to latest trends in manufacturing processes and materials.
4. Describe the fundamentals of automobile systems and the emerging trends of electric vehicles, hybrid electric vehicles and solar vehicles..

Course Outcomes:

After learning the course, the students should be able to:

1. **Identify** different materials and manufacturing processes for different applications.
2. **Apply** the fundamentals of thermodynamics and heat transfer to real life applications.
3. **Identify** the basic mechanical elements and the power transmission drives based on their applications.
4. **Comprehend** automotive system for the combustion vehicle, EV's, HEV's and solar vehicles..

Detailed Syllabus:

Unit	Description	Duration [Hrs]
I	**Materials and Manufacturing Engineering** Classification of materials, their properties and applications, phase change material (PCM), composite material. Introduction to manufacturing processes: classification and applications, sheet metal working and forging. Introduction to industry 4.0.	8
II	**Thermal Engineering** Introduction to thermodynamics: concept of a system, types of systems, energy interactions: heat and work, laws of thermodynamics, entropy, and modes of heat transfer, thermal resistance, insulation. Applications of hydraulic turbines, centrifugal pumps, compressor, household refrigerator, window and split air conditioner.	7
III	**Fundamentals of Design Engineering** Introduction to mechanical engineering: use of mechanical engineering in day to day life and its interdisciplinary use, Introduction to design thinking. Machine elements, power transmission drives and mechanisms.	8

IV	**Automotive Engineering** Classification of automobile, specifications of two wheeler, four wheeler and multi axle vehicles, types of chassis layout and drives, working of I.C. Engine, gear boxes, clutch, brakes, drive train system. Emission standards. Concept and environmental importance of electric vehicles (EV), construction and working of EV, hybrid electric vehicles (HEV) and solar vehicles, challenges and future scope of EV's and HEV's.	**7**
	Total	**30**

Text Books:

1. Basic Mechanical Engineering- Basant Agarwal and C. M. Agarwal, Wiley publication, First edition, (2008).
2. Engineering Thermodynamics- P. K. Nag, Tata McGraw-Hill publishing Co. Ltd., Sixth edition, (2017).
3. Heat and mass transfer- R. K. Rajput, S Chand publication, Revised edition, (2007).
4. A Textbook of Automobile Engineering- Khalil U Siddiqui, New Age International Publishers, Fifth edition, (2012).

Reference Books:

1. Design of Machine Elements- V. B. Bhandari, Tata McGraw-Hill publishing Co. Ltd., Fourth edition, (2017).
2. Theory of Machine- S. S. Ratan, Tata McGraw-Hill publishing Co. Ltd., Fifth edition, (2019).
3. Thermal Engineering- Arora and Domkunwar, Dhanpat Rai and Sons, Sixth edition, (2013).
4. Elements of Workshop Technology- Hajra, Chaudhari Volume I, Media Promoters and Publishers, Mumbai, Fourteenth edition, (2008).
5. Elements of Workshop Technology- Hajra, Chaudhari Volume II, Media Promoters and Publishers, Mumbai, Twelfth edition, (2007).
6. Handbook of Industry 4.0 and SMART Systems- Diego Galar Pascual, Pasquale Daponte, Uday Kumar, CRC Press, First edition, (2019).
7. Industry 4.0 - Dr. Bhushan Kelkar, Neuflex Publication, First edition, (2019).
8. Automobile Engineering Volume I- Dr. Kripal Singh, Standard Publishers Distributors, Eleventh edition, (2008).
9. Automation, production systems computer integrated manufacturing- Mikell Groover, Pearson Publication, Fourth edition, (2014).
10. Automotive Electrical Equipment- P. L. Kohli, McGraw Hill Education (India) Pvt Ltd., First edition, (2001).

Program:	B. Tech. CSE (AI & ML)					Semester:	III
Course :	OE-DSE- Neural Network and Fuzzy Logic Control (Offered by AS&H)					Code :	BSH23OE06
Credits	Teaching Scheme (Hrs./Week)			Evaluation Scheme and Marks			
	Lecture	Practical	Tutorial	FA		SA	Total
				FA1	FA2		
2	2	-	-	10	10	30	50

Prior Knowledge: Nil

Course Objectives:

This course aims to enable students to get acquainted with,

1. Knowledge of Neural Networks and its use for controlling real-time systems.
2. Knowledge about fuzzy set theory to solve various engineering problems.

Course Outcomes:

After learning the course, the students will be able to:

1. Compute feedforward Artificial Neural Network output using basic concepts related to Artificial Neural Network.
2. **Apply** backpropagation and optimizer algorithms to update weights of Neural Networks and application-based problem-solving.
3. Find fuzzification and defuzzification of crisp function using basic Fuzzy set theory concepts.
4. **Apply** a fuzzy logic control system to handle uncertainty and some engineering problems.

Detailed Syllabus

Unit	Description	Duration [Hrs]
I	**Architecture of Neural Network:** Introduction, Biological Neural Network, Artificial Neural Network The architecture of Artificial Neural Networks- Bias, Activation Function, Learning Methods, Learning rules, Types of Neural Networks: Single-layer, multi-layer, feed-forward, and recurrent neural networks.	7
II	**Neural Networks For Control:** Loss function, Weight initialization, Optimizers algorithms, Backpropagation Algorithm, Associative Memory Networks and their types, Discrete-time hop field networks	8
III	**Fundamental of Fuzzy Logic:** Fundamental of Fuzzy Logic: Classical sets, Fuzzy Sets, Membership function, Cardinality of fuzzy set, Fuzzy complement, Fuzzy Composition, properties and operation on Fuzzy sets, Fuzzy Relation, Fuzzification, Defuzzification	7
IV	**Fuzzy Logic Control:** Fuzzy Rule, Decision-making Logic, Linguistic variables Fuzzy Inference System: Mamdani FIS, Sugeno FIS Design of fuzzy controller, fuzzy optimization, applications of FIS to some real-life problem	8
	Total	30

Text Books:

1. Kosko, B, "Neural Networks and Fuzzy Systems: A Dynamical Approach to Machine Intelligence", Prentice Hall, NewDelhi, 2004.
2. Ross T. J., "Fuzzy logic with engineering applications (Vol. 2)", New York: Wiley, 2004, ISBN: 9783030375478

Reference Books:

1. Jack M. Zurada, "Introduction to Artificial Neural Systems," PWS Publishing Co., Boston, 2002.
2. Zimmerman H.J., "Fuzzy set theory and its Applications," Kluwer Academic Publishers Dordrecht, 2001.
3. Driankov, Hellendroonb, "Introduction to fuzzy control," Narosa Publishers,2001.
4. G Klir, B Yuan, "Fuzzy sets and fuzzy logic: Theory and application," PHI, ISBN:
5. LauranceFausett, Englewood Cliffs, N.J., "Fundamentals of Neural Networks," PearsonEducation, New Delhi, 2008.
6. B Yegnanarayana: Artificial Neural Networks for pattern recognition, PHI Learning Pvt. Ltd., 14-Jan-2009

e-sources:

1. Online course "Fuzzy Logic and Neural Network" by Prof. Dilip Kumar Pratihar,
2. IIT Kharagpur. https://nptel.ac.in/courses/127/105/127105006/

Program:	B. Tech. CSE (AI & ML)					Semester:	III
Course :	OE- DSE - Total Quality Management (Offered by Civil Department)					Code:	BCI23OE02
Credits	Teaching Scheme (Hrs./Week)			Evaluation Scheme and Marks			
	Lecture	Practical	Tutorial	FA		SA	Total
				FA1	FA2		
2	2	-	-	10	10	30	50

Prior Knowledge Basic definitions of Quality and importance of Quality in industry for safety and durability.

Course Objectives:

After Completing this course, student will have adequate background :

1. To understand the importance of Quality in construction.
2. To understand the need of Total Quality management & it's tools.
3. To understand role of ISO in quality management.

Course Outcomes:

After learning the course, the students should be able to:

1. Articulate quality and quality ideas as presented by many gurus and philosophers after learning.
2. Illustrate different quality control tools.
3. Apply ISO concepts and the cost of quality to quality assurance.
4. Analyze various techniques of TQM.

Detailed Syllabus

Unit	Description	Duration [Hrs]
I	**Unit I: Quality in Construction** a) Quality – Various definitions and interpretation. Importance of quality on a project in the context of global challenges, Factors affecting quality, Reasons for poor quality & measures to overcome, Contribution of various Quality Gurus (Juran, Deming, Crossby, Ishikawa), b) Evolution of TQM- QC, TQC, QA, QMS, TQM.	7
II	**Unit II: TQM, Six Sigma and QC tools** a) TQM – Necessity, advantages, Quality Function Deployment (QFD), b) Six sigma – Importance, levels, Application of 6 Sigma, c) Implementation of 7 QC tools through case study.	8
III	**Unit III: Cost of Quality and ISO** a) Categories of cost of Quality, b) Study of ISO 9001 principles., Quality manual – Importance, contents, documentation, Corrective and Preventive actions, Conformity and NC reports.	7
IV	**Unit IV: Techniques in TQM Implementation** a) Benchmarking in TQM, Kaizen in TQM, b) '5-S_techniques, Zero Defects, c) Quality Circle Concept and applications through Quality Circle Formation.	8
	Total	**30**

Text Books:

1. Total Quality Management-- Dr. Gunmala Suri and Dr. Puja Chhabra Sharma—Biztantra.
2. Quality Control and Total Quality Management by P.L.Jain- Tata McGraw Hill Publ.
3. Total Quality Management - Dr. S.Rajaram and Dr. M. Sivakumar—Biztantra.
4. Total Engineering Quality Management – Sunil Sharma – Macmillan India Ltd.

Reference Books:

1. Juran‘s Quality Handbook – Juran Publication. (2016 Edition)
2. Management –Principal, process and practices by Bhat – Oxford University Press.(2008)
3. Financial management by Shrivastava- Oxford University Press (6th Edition 2022)
4. Management Information Systems – Gordon B. Davis, Margrethe H. Olson – Tata McGraw Hill Publ. Co. (2022)
5. Total Project Management – The Indian Context - P.K.Joy Macmillan India Ltd.

Program:	B. Tech. CSE (AI & ML)					Semester:	III
Course :	OE-DSE- Building Services and Maintenance (Offered by Civil Department)					Code:	BCI23OE03
Credits	Teaching Scheme (Hrs./Week)			Evaluation Scheme and Marks			
	Lecture	Practical	Tutorial	FA		SA	Total
				FA1	FA2		
2	2	-	-	10	10	30	50

Prior Knowledge: NA

Course Objectives:

This course aims at enabling students,

1. To impart knowledge about the building services
2. To examine the purpose and type of building maintenance.

Course Outcomes:

After learning the course, the students should be able to:

1. Understand different building services provisions.
2. Interpret the fundamental concepts relevant to functional requirement of building.
3. Relate the knowledge of Acoustic and Fire Protection.
4. Choose diverse maintenance methodologies applicable to building and infrastructure services.

Detailed Syllabus

Unit	Description	Duration [Hrs]
I	**Introduction to Building Services:** Definitions, Objective and uses of services, different types of building, Classification of based on Occupancy, FSI, Carpet area, built-up area, Standard of Accommodation, Classification of building services, Types of services and selection of appropriate services for given project**,** case studies.	7
II	**Escalator and Plumbing-** Classification of different types of escalators, Lift codes and Rules, Design Features of Escalator, Plumbing- Common Sanitary Fixtures, Layout of Sanitary Fixtures, Water Pipe Sizing in Buildings, Building Services Detailing, Rain Water Harvesting, Sanitation in buildings.	8
III	**Acoustics and Ventilation-** Material properties, acoustical design of assembly halls and buildings, noise and its control, measuring equipment, Ventilation- Ventilation systems, health and comfort ventilation, Fire protection and National Building Codes 2016- requirements, equipment and their applications, security systems.	8
IV	**Building Maintenance:** Role of maintenance in durability and serviceability of structures, Economic aspects of maintenance, Different types of maintenance and audits process. **Infrastructures services:** Different types of structures, infrastructure services, case studies.	7
	Total	**30**

Text Books: 1. Building Construction Dr. B. C. Punmia Laxmi Publications (P) Ltd., New Delhi 2. Building Construction P. C. Varghese PHI Learning (P) Ltd., New Delhi 3. Building repair and Maintenance Management P. S. Gahlot CBS Publishers & Distribution(P) Ltd.
Reference Books: 1. Building Science & Planning by S.V. Deodhar, Khanna Publishers. 2. Design and Practical Hand Book on Plumbing by C.R Mohan, VivekAnand, Standard Publishers Distributors. 3. Hand book of Designing and Installation of Services in High Rise Building Complexes, by V.K. Jain, Khanna Publ

Program:	**B. Tech. CSE (AI & ML)**					**Semester:**	**III**
Course :	**Biology for Engineers (**Offered by E&TC)					**Code :**	**BET23OE03**
Credit	**Teaching Scheme**			**Evaluation Scheme**			
	Lecture	**Practical**	**Tutorial**	**FA**		**SA**	**Total**
				FA1	**FA2**		
2	2	-	-	10	10	30	50

Pre-requisite: Basics of Human Anatomy and physiology, Basics of Electronics Engineering

Objectives:

1. To introduce biological engineering principles, procedures needed to solve real-world problems.
2. To provide an overview of human anatomy and physiology in order to support biomedical engineering solutions.
3. To introduce biomedical sensors, signal processing and diagnostic systems.
4. To provide a basic knowledge of the applications of biological systems in relevant industries.

Outcomes:

After completing the course the students should be able :

1. to understand basics of human physiological system and its cell functioning
2. to understand human immune system and significance of microbiology
3. to map role of biology in designing industrial applications
4. to understand biomedical sensors, its interfacing and related to measurement systems.

Detailed Syllabus:

Unit	**Description**	**Duration (Hrs)**
1.	**CELL PHYSIOLOGY :** Introduction to the cell biology – Cell size and shape - Chemical composition - Classification of cell and its properties, Cell cycle; Cell signaling, Transport across cell membrane; Introduction to Human physiology – Circulatory system - Respiratory system - Excretory system - Nervous system.	08
2.	**IMMUNOLOGICAL SCIENCE:** Immune system and its types; Functional properties of antibodies; Helper T cells and T cell activation; Importance of Microbiology.	07
3.	**BIOLOGY AND ITS INDUSTRIAL APPLICATION:** Introduction : Bioreactors, biocontrol, biofilters, biopolymers, bioenergy, biomaterials, biochips, Nano-Biomolecules and its various types: Principles and Application of Biosensor; Basics of Biochips – Bio fertilizer – Bioinformatics – Bio fuel.	07
4.	**INTRODUCTION TO BIOMEDICAL INSTRUMENTATION:** Need and Challenges in measurement of the parameters in living systems, Source of bioelectric potential: Resting and action potential, propagation of action potential, depolarization and re-polarization. Introduction to few important bio-potential such as Electrocardiogram(ECG), Electroencephalogram(EEG) and Electromygram (EMG).	08
	Total	**30**

Text Books:

1. Dr. Sohini Singh and Dr. Tanu Allen, “Biology for Engineers”, Vayu Education Of India, New Delhi, 2014.
2. Leslie Cromwell, Fred J. Weibell, Erich A. Pfeiffer, Biomedical Instrumentation and Measurements, 2nd edition,Prentice Hall of India

Reference Books:

1. Arthur T. Johnson, "Biology for Engineers" CRC Press, 2011.
2. Goldsby RA, Kindt TK, Osborne BA and Kuby J (2003) Immunology, 5th Edition, W.H. Freeman and Company, New York.
3. Cell Biology and Genetics (Biology: The unity and diversity of life Volume I), Cecie Starr, Ralph Taggart, Christine Evers and Lisa Starr, Cengage Learning, 2008
4. Biotechnology Expanding horizon, B.D. Singh, Kalyani Publishers, 2012
5. John G. Webster(Editor), Amit J. Nimunkar (Editor), Medical Instrumentation: Application and Design,5th edition Wil publication
6. Joseph H. Carr, John M. Brown, Introduction to Biomedical equipment Technology, 4edition , Pearson publication

Program:	B. Tech. CSE (AI & ML)					Semester:	III
Course :	OE-DSE- Industry 4.0 (Open Elective Department Specific)					Code :	BME23OE04
Credits	Teaching Scheme (Hrs./Week)			Evaluation Scheme and Marks			
	Lecture	Practical	Tutorial	FA		SA	Total
				FA1	FA2		
2	2	-	-	10	10	30	50

Prior knowledge of Basic programming skills, Mathematical skills i**s essential.**

Course Objectives:

This course aims at enabling students,

1. To introduce revolutions of manufacturing industry
2. To introduce technological advancement in modern manufacturing industries
3. To Introduce concepts of smart manufacturing, emphasizing Industry 4.0 in manufacturing industries

Course Outcomes:

After learning the course, the students should be able to:

1. Correlate the recent manufacturing trends and technological pillars of Industry 4.0.
2. Apply pillars of Industry 4.0 to the manufacturing industry.
3. Adapt the changes in existing manufacturing practices and relate the role of industrial robotics and sensors.

Detailed Syllabus:

Unit	Description	Duration [Hrs]
I	**Industrial revolution and current state of industry:** Overview of industrial revolution, Introduction to Automation, hard automation, soft automation, classification of production system, adaptive control, overview of terminologies like CAD, CAM, CAE, CAPP etc.	06
II	**Introduction to Industry 4.0:** Introduction to industry 4.0, need for Industry 4.0, Framework for Industry 4.0, technological pillars in industrial 4.0, applications, challenges and scope for industry 4.0	08
III	**Technological developments in Industry 4.0:** Introduction to Smart Manufacturing, overview of big data and analytic techniques, cyber security, Internet of things (IoT), Industrial Internet of things (IIoT), Cloud computing, artificial intelligence.	08
IV	**Robotics and Sensors:** Introduction to technological components of Robot, classification of sensors and its applications in Manufacturing industry, Role of robots in Industry 4.0, Internet of Robotic Things, Cloud Robotics, and Cognitive Architecture for Cyber-Physical Robotics	08
	Total	30

Text Books: 1. Nine pillars of technologies for Industry 4.0, W. Leong, IET publishers, 2020 2. Industry 4.0, A. Gilchrist, Apress Publication, 2016 3. Industrial Automation: Hands On, Lamb, Frank. , McGraw-Hill Professional, 2013
Reference Books: 1. C. Schröder, The Challenges of Industry 4.0 for Small and Medium-sized Enterprises, 2021. 2. Chua C. K., Leong K. F., Lim C. S., Rapid Prototyping, World Scientific, 2012. 3. A. Nayyar and A. Kumar, A Roadmap to Industry 4.0: Smart Production, Sharp Business and Sustainable Development-Springer International Publishing, 2020. https://doi.org/10.1007/978-3-030-14544-6 4. K. Kumar, D. Zindani, J. P. Davim, Industry 4.0: Developments towards the Fourth Industrial Revolution, Springer Singapore, 2019.
E-sources: **1.** https://nptel.ac.in/courses/108105063

Program :	B. Tech. CSE (AI & ML)					Semester:	III
Course :	**OE-DSE- Energy Storage Management (**offered by Mechanical Department)					Code :	BME23OE05
Credits	Teaching Scheme (Hrs./Week)			Evaluation Scheme and Marks			
	Lecture	Practical	Tutorial	FA		SA	Total
				FA1	FA2		
2	2	-	-	10	10	30	50

Prior knowledge of Fundamentals of Engineering and basic sciences **is essential.**

Course Objectives:

This course aims at enabling students,

1. To explore energy storage systems' fundamentals, technologies, and applications.
2. To enable students to understand the principles of energy storage systems design.
3. To provide a general awareness of energy storage audit, safety, and management.

Course Outcomes:

After learning the course, the students should be able to:

1. Differentiate and select energy storage systems based on performance and safety.
2. Select material and analyze the energy storage in thermochemical form.
3. Analyze energy storage system with batteries/supercapacitors/fuel cells.
4. Perform audit of Energy storage systems

Detailed Syllabus:

Unit	Description	Duration [Hrs]
I	**Energy storage system:** Introduction, Need, Modes, Energy storage devices, Merits, and Demerits of different types of Storage, utilization, and system applications. **Performance:** Energy capture rate and efficiency, Discharge rate and efficiency, Dispatch ability and load flowing characteristics, Scale flexibility, durability, Cycle lifetime, mass.	8
II	**Thermochemical storage –** Types, Phase Change Materials (PCMs) and classifications, properties of the PCMs for different temperature ranges, Selection criteria of PCMs for heating and cooling in buildings, Application of PCM in energy storage. SHS mediums, Energy analysis of the latent heat storage based on different systems	7
III	**Energy Storage for transportation:** Mechanical Design and Packaging of Battery, Packs for Electric Vehicles, Advanced Battery-Assisted Quick Charger for Electric Vehicles, Charging Optimization Methods for Li-ion Batteries, Thermal run-away for battery systems, State of Charge and Health Estimation Over the Battery Lifespan, Recycling of Batteries from Electric Vehicles **Fuel cell**: Operational principle, types, hybrid fuel cell-battery systems, hybrid fuel cell-supercapacitor systems,	8
IV	**Energy Storage Management** Relationship between economic growth and energy use, Energy demand analysis in different economic sectors, increase energy conversion efficiencies by introducing energy storage, Energy audit of energy storage system, The carbon markets, Safety: Risks of fire, explosion, toxicity, Ease of materials, recycling, and recovery, Environmental consideration, and recycling	7

	Total	30

Text Books:

1. A. R. Pendse, "Energy Storage Science and Technology," SBS Publishers & Distributors Pvt. Ltd., New Delhi, (ISBN – 13:9789380090122), 2011.
2. Jiujun Zhang, Lei Zhang, Hansan Liu, Andy Sun, Ru-Shi Liu, "Electrochemical Technologies for Energy Storage and Conversion," John Wiley and Sons, 2012.
3. Lithium-Ion Batteries Basics and Applications by Reiner Korthauer, Springer.
4. Fuel cells from fundamentals to applications by Supramaniam Srinivasan, Springer.

Reference Books:

1. Frank S. Barnes and Jonah G. Levine, Large Energy Storage Systems Handbook (Mechanical and Aerospace Engineering Series), CRC press (2011)
2. Francisco Díaz-González, Andreas Sumper, Oriol Gomis-Bellmunt," Energy Storage in Power Systems" Wiley Publication, ISBN: 978-1-118-97130-7, Mar 2016.
3. Pistoia, Gianfranco, and Boryann Liaw. The behavior of Lithium-Ion Batteries in Electric Vehicles: Battery Health, Performance, Safety, and Cost. Springer International Publishing AG, 2018.
4. E. Lipman, A. Z. Weber, Fuel Cells and Hydrogen Production, A Volume in the Encyclopedia of Sustainability Scien and Technology, Second Edition, Springer reference.
5. Handbook of Energy Audit by Sonal Desai Publisher Tata McGraw Hill.

Program :	B. Tech. CSE (AI & ML)					Semester:	III
Course :	Business Studies for Engineers (Offered by Department of Applied Sciences & Humanities)					Code :	BSH23EM01
Credits	Teaching Scheme (Hrs./Week)			Evaluation Scheme and Marks			
	Lecture	Practical	Tutorial	FA		SA	Total
				FA1	FA2		
2	2	-	-	10	10	30	50

Prior knowledge : NIL

Course Objectives:

This course aims at enabling students:

1. To help the students to gain understanding of various perspectives in the field of Strategic Management
2. To enable the students to pursue modern management practices in business.
3. To provide the students an understanding about tools and techniques of economic principles in business management.

Course Outcomes:

After learning the course, the students will be able to :

1. Explain the concept of Management and Strategic Management processes through case studies.
2. Illustrate the Management Trends and Practices implied in Global Work Culture.
3. Identify the role of economic variables in the business economy.
4. Analyze the business expansion strategies abroad and key issues related to their operations.

Detailed Syllabus:

Unit	Description	Duration [Hrs]
I	**Business Finance:** Concept of Business Finance, Objective and Scope, Significance of Finance, principles of business finance, Finance Function, Business finance v/s Corporate finance, Source of Finance. Meaning of financial planning, steps in financial planning, significance of financial planning, essential features of a good financial plan, Personal financial planning, Types of financial plan.	7
II	**Capitalization :** Introduction to capitalization, Amount of Capitalization, Over Capitalization, Under Capitalization, Venture Capital - Meaning of Venture Capital, Method of Venture Financing ,Venture Capital Funds, Policies and Procedures adopted by Venture Capitalists, Venture Capital in India, Guidelines for Venture Funds.	7
III	**Financial Markets, Institutions and Instruments:** Introductions to Financial Markets: Nature, Functions and Types of Financial markets, Different Financial Instruments, Sources of financing: Shares, Debentures, Term Loans, Retained Earnings, Public Deposits, Bonds, Trade Credit, Mutual Funds - Definition, Types of Mutual Funds, Significance of Mutual Funds; Case Studies on Financial Markets.	8

IV	**Constituents of the Financial System and Regulatory Institutions :** Classification: Fund Based, Non Fund Based and Modern Services, Hire Purchasing, Leasing: Lease Financing - Essential Elements of Leasing, Types of Leases, Merits and Demerits of Lease Financing, Merchant Banking - Role; Functions of Merchant Banking, Factoring. Debt management, Portfolio Management. RBI - Organization, objectives, role and functions, monetary policy of RBI, NABARD, SEBI - Organization and Objectives	**8**
	Total	**30**

Text Books:

1. Srivastava, R.M. Essentials of Business Finance, Himalaya Publishing House, Kalyani Publications

Reference Books:

1. Gordon, E. & Natarajan, K. Financial Markets and Institutions, Himalaya Publishing House.
2. Khan and Jain, Financial Management, Tata McGraw Hill, 2008
3. Singh, Preeti. Investment Management. Himalaya Publishing House,
4. Kale, N.G. Business Organization. Manisha Publications.

e-sources:

1. https://www.youtube.com/watch?v=TgF2XvjquUU&list=PLLy_2iUCG87CXY2B6fPex1SOIqxzzD5Wj
2. https://www.youtube.com/watch?v=CCQwz_Gwo6o
3. https://www.youtube.com/watch?v=OT5RdoJAkhY&list=PLPjSqITyvDeUTeAOGhip_ubjN3y8oqT13

Program:	B. Tech. CSE (AI & ML)				Semester:		III
Course:	Object-Oriented Programming Laboratory				Code:		BCS23PC04
Credits	Teaching Scheme (Hrs./Week)			Evaluation Scheme and Marks			
	Theory	Practical	Tutorial	TW	OR	PR	Total
01	-	02	-	25	25	–	50

Prior Knowledge of Computer Programming and Problem Solving is essential.

Course Objectives:

1. To develop solutions by using various OOP features for various applications.
2. To strengthen problem-solving ability by using the characteristics of an object-oriented approach.

Course Outcomes:

After learning the course, students will be able to:

1. Apply various object-oriented concepts for problem-solving.
2. Apply inheritance and polymorphism to write object-oriented programs.
3. Develop the solutions using generic templates, exception handling, and STL.
4. Write programs to perform file-handling operations.

Guidelines for Laboratory Conduction:

1. Instructor is expected to teach Object oriented programming concepts before conducting assignments.
2. Each student must perform at least 10 assignments.
3. Assignments should be implemented in C++ language.
4. Operating System recommended: - 64-bit Open source Linux or its derivative.
5. Programming tools recommended: - G++/GCC, Eclipse,VS Code, QT creator.

Assignment No.	Suggested List of Assignments (All assignments are compulsory)
1.	**Class, Object, Member Function and Access specifier:** Write a program Illustrating Class Declarations, Definition, and Accessing Class Members. a. Write a C++ program to implement a class called Circle that has private member variables for radius. Include member functions to calculate the circle's area and circumference.
2.	Write a C++ Program to Check Characters are Uppercase, Lowercase, Digit or SpecialCharacter.
3.	**Constructor, its types and Destructor** Write a program to illustrate default constructor, parameterized constructor and copy constructors. Program should illustrate the order of execution of constructors and destructors when a new class is derived from more than one base class.
4.	**Friend Class, friend Function, Static data Members, Static Function, Inline Function:** a. Write a Program Using following concepts: b. a. Friend Class, b. friend Function, c. Static data Members, d. Static Function, e. Inline Function

5.	**Inheritance and its Types** Write C++ programs that illustrate how the following forms of inheritance are supported: a. Single inheritance b. Multiple inheritance c. Multilevel inheritance d. Hierarchical inheritance.
6.	**Poly morphism : Operator Overloading, Virtual Function** 1. Implement a class Complex, which represents the Complex Number data type. Implement the following operations: a. Constructor (including a default constructor which creates the complex number 0+0i). b. Overloaded operator + to add two complex numbers. c. Overloaded operator * to multiply two complex numbers. d. Overloaded << and >> to print and read Complex Numbers. 2. Write a C++ program to implement a class called Shape with virtual member functions for calculating area and perimeter. Derive classes such as Circle, Rectangle, and Triangle from the Shape class and override virtual functions accordingly.
7.	**ADT Using Class** Write a C++ program to implement the matrix ADT using a class. The operations supported by this ADT are: a. Reading a matrix. b. Addition of matrices. c. Printing a matrix. d. Subtraction of matrices. e. Multiplication of matrices.
8.	**Generic Programming Class Template and Function Template:** Write a class template to represent a generic vector. Include member functions to perform the following tasks: a. To create the vector. b. To modify the value of a given element. c. To multiply the vector by a scalar value. d. To display the vector in the form (10, 20, 30)
9.	**Exception Handling** 1. Write a C++ to handle the exception if account balance is less than minimum balance.
10.	**File Handling** 1. Write a program to maintain an employee database in a binary file with employee information such as empId, name, age, department, post and salary. Write function for adding new records, displaying all records, searching for a particular employee, updating employee salary and post.
11.	**Standard Template Library: Stack** Write C++ program using STL to add binary numbers (assume one bit as one number); use STL stack.
12.	**Standard Template Library: Stack and Queue** Write a C++ program using STL to perform various stack and queue operations.
13.	**Standard Template Library: Map** Write C++ program using STL map for managing Person Records (Name, birth date, telephone

	no). Perform operations – add, display, search, delete, and update.

Text Books:

1 Robert Lafore, "Object Oriented Programming in C++", person Publishing, Forth Edition, 2002, ISBN: 0672323087.

2 Herbert Schildt, "C++: The Complete Reference", McGraw Hill Education, 4th Edition, July 2017, ISBN-10 : 007053246X (ISBN-13 : 978-0070532465).

Reference Books:

1. Bjarne Stroustrup, "The C++ Programming language" , Pearson Education , Third edition, 2008, ISBN 9780201889543.

Program:	B. Tech. CSE (AI & ML)				Semester:	III	
Course:	Data Structures Laboratory				Code:	BCS23PC05	
Credits	Teaching Scheme (Hrs./Week)			Evaluation Scheme and Marks			
	Theory	Practical	Tutorial	TW	OR	PR	Total
02	-	04	-	50	–	50	100

Prior Knowledge of Data structures is essential.

Course Objectives:

1. To implement linear and nonlinear data structures using C++.
2. To apply various sorting and searching techniques.
3. To analyze performance measures of algorithms.

Course Outcomes:

1. Analyze the performance of various algorithms based on time & space complexity.
2. Select appropriate data structure and demonstrate a working solution for a given problem.
3. Develop programs, using linear and nonlinear data structures, to solve various engineering problems.
4. Implement hashing, sorting, and searching techniques to solve real life problems.

Guidelines for Laboratory Conduction:

1. The instructor is expected to conduct a minimum 10 assignments from the suggested list.
2. The instructor is expected to address at least one assignment related to each concept outlined in the theory syllabus.
3. Encourage students for appropriate use of Hungarian notation, proper indentation and comments.
4. Use of open source software is to be encouraged.
5. Assignments should be implemented in C++ language.
6. Operating System recommended: - 64-bit Open source Linux or its derivative.
   Programming tools recommended: - G++/GCC, Eclipse, QT creator

Assignment No.	Suggested List of Assignments
1.	For a given number of integers, select a data structure to rearrange it such that every second element becomes greater than its left and right elements. Assume no duplicate elements are present in the list of integers.
2.	The two strings, X and Y, are anagrams if by rearranging X's letters, we can get Y using all the original letters of X exactly once. For example, all these pairs are anagrams as LHS can be rearranged to RHS and vice-versa. Select the appropriate data structure using which the anagrams should be grouped together.

3.	Write a Program for magic square. A magic square is a square matrix of size n x n, where the integers from 1 to n^2 are arranged in such a way that the sum of each row, each column, and both main diagonals is the same. In this problem, you are tasked with creating a program that generates and verifies a magic square for a given odd value of n.
4.	The Department of AI&ML has a student's club named 'ARTIMAS'. Students of second and third year of department can be granted membership on request. Similarly one may cancel the membership of a club. First node is reserved for the president of the club and the last node is reserved for the secretary of the club. Write C++ program to maintain club member‘s information using singly linked lists. Store student PRN and Name. Write functions to: a) Add and delete the members as well as president or even secretary. b) Compute total number of members of club c) Display members d) Two linked lists exist for two divisions. Concatenate two lists.
5.	The ticket booking system of Cinemax theater has to be implemented using C++ program. There are ‘m’ rows and ‘n’ seats in each row. Doubly circular linked list has to be maintained to keep track of free seats at rows. Assume some random booking to start with. Use an array to store pointers (Head pointer) to each row. Write the functions for the following: a) The list of available seats is to be displayed b) The seats are to be booked c) The booking can be canceled.
6.	Implement C++ program for expression conversion to evaluate an infix to postfix using appropriate data structure considering the given constraints as below: 1. Operands and operators, both must be single characters. 2. Input Postfix expression must be in a desired format. 3. Only '+', '-', '*' and '/ ' operators are expected.
7.	Implement a simple calculator program that continuously takes mathematical expressions from the user and evaluates them using the best suitable data structure. The program should support basic arithmetic operations (+, -, *, /) and handle invalid expressions gracefully.
8.	Queues are frequently used in computer programming, and a typical example is the creation of a job queue by an operating system. If the operating system does not use priorities, then the jobs are processed in the order they enter the system. Write C++ program for simulating job queue. Write functions to add jobs and delete jobs from the queue.
9.	Pizza parlor accepting maximum M orders. Orders are served on a first come first served basis. Order once placed cannot be canceled. Write C++ program to simulate the system using a circular queue using arrays.
10.	A book consists of chapters, chapters consist of sections and sections consist of subsections. Construct a tree and print the nodes. Find the time and space requirements of your method.
11.	Convert a given binary tree into a threaded binary tree. Analyze time and space complexity of the algorithm.

12.	You are given a sorted list of integers. Write a program to implement a binary search algorithm to find a target integer in the list. Evaluate the time & space complexity of the searching techniques. Input: The input will consist of an integer n, the number of elements in the list ($1 \leq n$) where integers will be separated by n space, representing the list of integers. Output: Print the index of the target integer in the list. If the target integer is not present in the list, print -1.
13.	You are given an undirected graph represented as an adjacency list.Implement a function that performs Breadth-First Search and Depth-First Search on the given graph, starting from the specified vertex. The function should return a list representing the order in which vertices are visited during the traversal.
14.	You have a business with several offices; you want to lease phone lines to connect them up with each other; and the phone company charges different amounts of money to connect different pairs of cities. You want a set of lines that connects all your offices with a minimum total cost. Solve the problem by suggesting appropriate data structure's algorithm.
15.	Consider a real-time scenario where an e-commerce platform manages a vast catalog of products from various categories. The platform needs to efficiently sort and present products to users based on different criteria. Write a program - 1. To display products sorted by their prices to help users find items within their preferred price range. (Quick sort / Merge sort) 2. To arrange products by their popularity metrics.(Insertion sort / Selection sort) Analyze the execution time and memory usage of each sorting algorithm to determine their efficiency in handling such large product catalogs.
16.	The Employee Authentication System needs to store employee information, including their usernames and hashed passwords.Use a hash table to store employee records where each record contains a username and the corresponding hashed password.Hash the entered password and compare it with the stored hashed password in the hash table for authentication.

Text Books:

1. Horowitz and Sahani, "Fundamentals of Data Structures in C++", University Press, ISBN 10:0716782928 /ISBN 13: 9780716782926.
2. Brassard & Bratley, "Fundamentals of Algorithms", Prentice Hall India/Pearson Education, ISBN 13-9788120311312.

Reference Books:

1. Horowitz, Sahani and Rajshekaran, "Fundamentals of Computer Algorithms", University Press, ISBN 13: 9788175152571.
2. Yedidyah Langsam, Moshe J Augenstein, Aron M Tenenbaum, "Data Structures using C and C++", Pearson Education, ISBN 81-317-0328-2.

Program:	B. Tech. CSE (AI & ML)					Semester:	III
Course :	VEC-I- Universal Human Values					Code:	BSH23VE01
Credits	Teaching Scheme (Hrs./Week)			Evaluation Scheme and Marks			
	Lecture	Practical	Tutorial	FA		SA	Total
				FA1	FA2		
2	2	-	-	20	30	-	50

Course Objectives:

This course aims at enabling students,

1. To appreciate the essential complementarity between 'VALUES' and 'SKILLS' to ensure sustained happiness and prosperity, which are the core aspirations of all human beings.
2. To facilitate the development of a holistic perspective among students to lead their personal and professional lives in an ethical way.
3. To highlight plausible implications of such a holistic understanding in terms of ethical human conduct, trustful and mutually fulfilling human behavior, and mutually enriching interaction with nature.

Course Outcomes:

After learning the course, the students should be able to:

1. **Explain** the relevance of 'Universal Human Values'
2. **Develop** an understanding about human being as coexistence of 'Self' & 'Body'
3. **Apply** the sense of harmony in family and society
4. **Take part** in ensuring coexistence with nature by integrating Universal Human Values into personal and professional lives.

Detailed Syllabus:

Unit	Description	Duration [Hrs]
I	**Introduction to Value Education:** ● Understanding Value Education ● Self-exploration as the Process for Value Education ● Continuous Happiness and Prosperity ● Right Understanding ● Current Scenario ● Method to fulfill the Basic Human Aspirations	3
	Practice Session: ● Sharing about Oneself ● Exploring Human Consciousness ● Exploring Natural Acceptance **Experiential Learning** ● Seva Activity	3
II	**Harmony in the Human Being:** ● Human being: the Co-existence of the Self and the Body ● Needs of the Self and the Body ● The Body as an Instrument of the Self ● Understanding Harmony in the Self ● Harmony of the Self with the Body ● Programme to Ensure Self-Regulation and Health	4

	Practice Session: ● Exploring the Difference between Needs of Self and Body ● Exploring Sources of Imagination in the Self ● Exploring Harmony of Self with the Body **Experiential Learning** ● Seva Activity ● Health Awareness Programme	**4**
III	**Harmony in the Family and in Society:** ● Harmony in the Family ● 'Trust'—the Foundational Value in Relationship ● 'Respect'—the Right Evaluation ● Other feelings (Values) in Human-to-Human Relationship ● Understanding Harmony in Society ● Vision for the Universal Human Order ● Five Dimensions of Human Order	**4**
	Practice Session: ● Exploring the Feeling of Trust ● Exploring the Feeling of Respect and Exploring Systems to Fulfil Human Goal **Experiential Learning** ● Seva Activity	**4**
IV	**Harmony in Nature/Existence:** ● Understanding Harmony in Nature, ● Realising Existence as Coexistence at All Levels ● The Holistic Perception of Harmony in Existence **Implications of Holistic Understanding: A Look at Professional Ethics:** ● Definitiveness of (Ethical) Human Conduct ● Humanistic Constitution and Universal Human Order ● Competence in Professional Ethics ● Holistic Technologies, Production Systems and Management Models-Typical Case Studies ● Strategies for Transitioning towards Value-Based Life and Profession.	**4**
	Practice Session: ● Exploring the Four Orders of Nature ● Exploring Co-existence in Existence ● Exploring Ethical Human Conduct ● Exploring Humanistic Models in Education ● Exploring Steps of Transition towards Universal Human Order **Experiential Learning Activity** ● Health Awareness Programme/Waste Management Programme (Hospital Waste/Pharmaceutical Industrial Waste/Reduce Plastic Waste / E-Waste Management)	**4**
	Total	**30**

Text Books:

1. R R Gaur, R Sangal, G P Bagaria, 2019, A Foundation Course in HUMAN VALUES and Professional Ethics-Presenting a universal approach to value education through self-exploration, Excel Books

Reference Books:

1. P.L. Dhar, RR Gaur, 1990, Science and Humanism, Commonwealth Publishers.
2. A. Nagaraj, 1999, JeevanVidya: EkParichaya, JeevanVidyaPrakashan, Amarkantak,
3. A. N. Tripathy, 2003, Human Values, New Age International Publishers.
4. E. G. Seebauer& Robert L. Berry, 2000, Fundamentals of Ethics for Scientists & Engineers , Oxford University Press
5. M. Govindrajran, S Natrajan& V.S. Senthil Kumar, Engineering Ethics and Human Values, Eastern Economy Edition, Prentice Hall of India Ltd.
6. B. P. Banerjee, 2005, Foundations of Ethics and Management, Excel Books.
7. B. L. Bajpai, 2004, Indian Ethos and Modern Management, New Royal Book Co., Lucknow. Reprinted 2008.

e-sources:

1. http://madhyasth-darshan.info/postulations/knowledge/knowledge-of-humane-conduct/
2. https://www.youtube.com/channel/UCQxWr5QB_eZUnwxSwxXEkQw
3. https://youtu.be/OgdNx0X923I

Program:	B. Tech. CSE (AI & ML)				Semester:	III	
Course:	Community Engineering Project				Code:	BCS23EL01	
Credits	Teaching Scheme (Hrs./Week)			Evaluation Scheme and Marks			
	Theory	Practical	Tutorial	TW	OR	PR	Total
02	-	04	-	50	50	–	100

Prior knowledge of Artificial Intelligence, Python and Machine Learning is essential.

Course Objectives:

1. To introduce the students to the community problems/needs.
2. To understand the complexity of the dataset and the size of the dataset.
3. To build Machine learning model to solve community problems

Course Objectives:

After learning the course, the students will be able to:

1. To study and understand the societal/community needs
2. To identify the inputs/data required to solve a specific society problem
3. To select and apply the relevant ML algorithm to solve a specific problem
4. To design and develop the prototype/product in line with social responsibility and ethics.

Guidelines for Students:

1. Students are required to identify at least one real life problem from group-A(Drinking Water/ Agriculture/Healthcare/Weather/Air Pollution) and group-B(Non conventional Energy/Risk Assessment and decision taking/Project Management/ Construction/Transportation/Finance, etc) and work in team for specific domain
2. The problem chosen will be relevant to the societal needs and students will undergo field visits (within 100 Km) under the guidance of the faculty for identifying the community requirements to improve the quality of living of the community at large.
3. Group of 4-5 students can form a team. After identifying and defining the problem, the time plan of activities and methodology will be framed using relevant scientific methods.
4. Field work: Two Field visits are compulsory
   Visit to a local area to document environmental asset: -River/Forest/Hill/Mountain
   - Visit to a local polluted site-Urban/Rural/Industrial/Agricultural
   - Study of common plants, insects, birds.
   - Study of simple ecosystems-pond, river, hill slopes, etc.
5. Each member of the team will prepare his/her own report highlighting the contribution majorly with brief abstract of the total work in the initial part of the document.
6. No two reports will be the same. The review will be based on the individual contribution.

List of Assignments (Perform any two Assignment one from group-A and one from group-B)

Assignment No.	Suggested List of Assignments
	Group A

1.	**Drinking Water Quality Prediction** Water quality can be measured with measuring the dissolved oxygen (DO), chemical oxygen demand (COD), biochemical oxygen demand (BOD), electrical conductivity (EC), pH, temperature, etc. Various Machine learning algorithms can studied for water quality prediction and classification. Similar to Dinking water prediction, students can also work on Water classification, quality monitoring and controlling for water purification which can used for agriculture and industrial use
2.	**Weather Forecasting** Traditional weather forecasting methods rely on complex physical models, while ML offer a more data-driven approach, identifying patterns and relationships in historical weather data to predict future conditions. By combining the strengths of traditional methods, ML techniques, and human expertise, meteorologists can achieve more accurate and timely forecast precise weather information.
3.	**Air Pollution:** Air pollution in the modern world is a matter of grave concern. Due to rapid expansion in commercial social and economic aspects, the pollutant concentrations in different parts of the world continue to increase and disrupt human life. Thus, monitoring the pollutant levels is of primary importance to keep the pollutant concentrations under control. Regular monitoring enables the authorities to take appropriate measures in case of high pollution. Machine Learning algorithms can be used to predict the Air pollution at different location with the help of conventional pollution monitoring stations.
4.	**Agriculture Crop Estimation** A large part of the world population has been engaged in agriculture and its related activities since ancient times. Farmers have usually relied on their own experiences to make any decision about agriculture or farm management. As agriculture is greatly affected by unpredictable weather conditions, it becomes difficult to take decisions with total confidence, and many a times farmers have to suffer great losses. If we could predict with certain accuracy, the future yield of crops, it would greatly help the farmers in deciding which crops to focus on, to obtain maximum profits. Students are expected to predict a crop yield using machine learning algorithms (decision trees, random forest, support vector regressor, gradient boosting) which can be used to predict the yields of major crops produced in the specific region in India. Historical production and meteorological data can be collected and processed for analysis and applying ML algorithms. The models were validated by testing on data they have not trained on.

5.	**Healthcare diagnostics:** A)Prediction of Diabetic: Objective of this project is to build a machine learning model to accurately predict whether or not the patients in the dataset have diabetes or not? Diabetes is a group of metabolic disorders in which there are high blood sugar levels over a prolonged period. Symptoms of high blood sugar include frequent urination, increased thirst, and increased hunger. If left untreated, diabetes can cause many complications. Acute complications can include diabetic ketoacidosis, hyperosmolar hyperglycemic state, or death. Serious long-term complications include cardiovascular disease, stroke, chronic kidney disease, foot ulcers, and damage to the eyes. The students are required to understand the need of various features to be collected and generated. Also study the on-line study the data set and predict the diabetes B) Prediction of cancer: breast cancer/brain cancer/Lung cancer Objective of this project is to build a machine learning model to accurately predict whether or not the patients in the dataset have cancer or not? If yes cancer stage? Ex1: Breast cancer is the most common cancer amongst women in the world. It accounts for 25% of all cancer cases, and affected over 2.1 Million people in 2015 alone. It starts when cells in the breast begin to grow out of control. These cells usually form tumors that can be seen via X-ray or felt as lumps in the breast area. The key challenge is to collect the cancer dataset (breast/Lung/brain, etc) and study its details and build a Machine learning model to classify tumors into malignant (cancerous) or benign(non cancerous). Also study the on-line data set
6.	**Study of Non conventional Energy systems:** Students are advised to visit non conventional energy systems like Solar plants and/or Wind Mills and study the entire system in terms of its efficiency and its comparison to conventional energy generation plants. They can study how the efficiency of the non conventional energy systems can be improved with the help of Machine learning algorithms.
Group B	
7.	**Risk Assessment and Decision-Making** AI may assist in analyzing the settlement of foundations, slope stability, and other geotechnical aspects. It also may predict risks and may provide solutions during different project stages, aiding decision-making.
8.	**Construction Waste Management** AI may help to optimize waste management by identifying smart materials and handling them efficiently. Expert monitoring may ensure cost optimization throughout the work system.
9.	**Project Management** AI-based project management tools may streamline tasks, enhance communication, and improve overall efficiency. Structural health monitoring, powered by AI, may ensure the safety and reliability of buildings and infrastructure.
10.	**Optimization and Productivity Enhancement:** AI may help to improve productivity by reducing idle time. For instance, it may optimize construction schedules, resource allocation, and task sequencing. ML algorithms may be able to predict maximum dry density and optimum moisture content in concrete, leading to better material utilization and cost savings.
11.	**Structural Health Monitoring (SHM):** AI techniques, such as machine learning, can be applied to detect damages in structures using sensory or image data. This may include identifying the location and extent of cracks, deformations, or other structural issues. By analyzing real-time data from sensors or cameras, engineers can predict when and where repairs are needed, ensuring the safety and longevity of buildings and infrastructure.

12.	**Transportation optimization** ML has already become a crucial prospect to delivery companies and other transportation enterprises. They use it to analyze data to identify patterns and even trends, making routes more efficient and even predicting potential problems to increase profitability.
13.	**Financial Services/Risk Assessment** Banks and other financial institutions in advanced countries are using ML for two key purposes: preventing fraud and identifying essential insights in data.
14.	**Construction Site Monitoring** Image recognition and AI algorithms may enable proper site monitoring. Safety aspects, dangerous conditions, and compliance with regulations can be assessed automatically. Real-time monitoring may help track progress, identify bottlenecks, and manage costs based on raw material market rates.

Reference Books:

1. Bishop M., C., Pattern Recognition and Machine Learning, Springer-Verlag (2011) 2nd Ed.
2. Mitchell M., T., Machine Learning, McGraw Hill (1997) 1st Edition, ISBN : 0070428077 9780070428072
3. Alpaydin E., Introduction to Machine Learning, MIT Press (2014) 3rd Edition.
4. Michie D., Spiegelhalter J. D., Taylor C. C., Campbell, J., Machine Learning, Neural and Statistical Classification. Overseas Press (1994).
5. Duda, Hart, Stork, Pattern Classification, Wiley, 2001

Course Syllabus

Second Year B.Tech.

Computer Science & Engineering

(Artificial Intelligence & Machine Learning)

Semester - IV

Program :	B. Tech. CSE (AI & ML)					Semester:	IV
Course :	Theory of Computation					Code:	BCS24C06
Credits	Teaching Scheme (Hrs./Week)			Evaluation Scheme and Marks			
	Lecture	Practical	Tutorial	FA		SA	Total
				FA1	FA2		
3	2	-	1	20	20	60	100

Prior knowledge of Computer Programming, Problem Solving and python is essential.

Course Objectives:

1. To recall and understand the basics of mathematical concepts, formal languages and machines.
2. To understand and design different computational models like finite automata, regular expression, push down automata, context free grammar, Turing machine for a given language.
3. To apply inter conversion between equivalent representations of a language.
4. To design appropriate computational models.

Course Outcomes :

After learning the course, students will be able to:

1. Construct a finite automaton to solve problems.
2. Design and Analyze regular expressions for the regular languages.
3. Analyze and Design Context Free Grammars and learn to simplify the grammar
4. Construct Pushdown Automaton model for the Context Free Language
5. Design and analyze Turing machines for formal languages.
6. Classify different classes of problems as Decidable, Undecidable , NP or P.

Detailed Syllabus:

Unit	Description	Duration [Hrs]
I	**Finite Automata:** Introduction to Finite Automata, Deterministic Finite Automata (DFA), Nondeterministic Finite Automata (NFA), Finite Automata with Epsilon Transition.Definition of Regular Language, Deterministic and Non deterministic FA(DFA and NFA) Moore and Mealy machines -Definition, models, inter-conversion. Case Study: Finite State Machine based vending machine	9
II	**Regular Expressions:** Regular Expressions, Finite Automata and Regular Expressions, Applications of Regular Expressions, Proving Languages Not to Be Regular, Closure Properties of Regular Languages, Equivalence with finite automata, Non regular Languages: The pumping lemma for regular languages.(Tool) Case Study: RE use in text search and replace operations	7

III	**Context Free Grammar & Languages:** Types of Grammar, Ambiguity in Grammars and Languages, Eliminating Useless Symbols, Computing the Generating and Reachable Symbols, Eliminating Epsilon Productions, Eliminating Unit Productions, Chomsky Normal Form, Closure Properties of Context Free, Applications of Context Free Grammars. Case Study- CFG for Palindromes and Parenthesis Matching,	**7**
IV	**Pushdown Automata:** Definition of the Pushdown Automaton, Instantaneous Description, Language of PDA, Acceptance by Final state, Acceptance by empty stack, Deterministic and non-deterministic PDA, Equivalence of PDA and CFG, CFG to PDA and PDA to CFG. Case Study : Top-Down Parsing, Bottom-up Parsing simulation showing use of PDA	**7**
V	**Turing Machine:** Turing Machines: Formal definition of a Turing machine. Variants of Turing Machines: Multi-tape Turing machines, Non-deterministic Turing machines, Halting problem of Turing Machine, Recursively Enumerable languages. Examples of Turing Machine	**8**
VI	**Decidability:** Decidable Languages: Decidable problems concerning regular languages, Decidable problems concerning context-free languages, Un-decidable Problems, **Reducibility:** Undecidable Problems that is recursively enumerable. A Simple Un-decidable. **Complexity Classes:** Time and Space Measures, The Class P, Examples of problems in P, The Class NP, Examples of problems in NP, P Problem Versus NP Problem, NP-completeness and hard Problems.	**6**
	Total	**45**

Text Books:

1. Peter Linz, "An Introduction to Formal Languages and Automata", 6th Edition, Jones & Bartlett Learning, 2016, ISBN : 9781284077247, 1284077241
2. John E. Hopcroft, Rajeev Motwani, Jeffrey D.Ullman, "Introduction to Automata Theory Languages and Computation", Pearson, 3rd Edition, 2008, ISBN-13: 978-8131720479.

Reference Books:

1. Daniel I.A. Cohen, "Introduction to Computer Theory", 2nd Edition , John Willy & Sons Inc., Sons Inc., 2000
2. Vivek Kulkarni "Theory of Computation", Oxford University Press, 2013, ISBN-13 978-0198084587
3. John C Martin, "Introduction to Languages and Theory of Computation", 4th Edition, Tata McGraw-Hill, 2011, ISBN: 978-81-317-6177-9

e-sources: NPTEL Video Courses:
1. https://nptel.ac.in/courses/106104148

Program :	B. Tech. CSE (AI & ML)					Semester:	IV
Course :	Database Management Systems					Code :	BCS24PC07
Credits	Teaching Scheme (Hrs./Week)			Evaluation Scheme and Marks			
	Lecture	Practical	Tutorial	FA		SA	Total
				FA1	FA2		
2	2	-	-	10	10	30	50

Prior Knowledge of: 1. Discrete Mathematics 2. Data Structures are essential.

Course Objectives:

1. To understand the fundamental concepts of database systems.
2. To learn Database query language.
3. To learn the concept of normalization, transaction, concurrency control and recovery techniques.
4. To learn unstructured data management.

Course Outcomes :

After learning the course, students will be able to:

1. Design ER and EER diagrams for real life applications.
2. Write queries and PL/SQL Code blocks.
3. Design schema in appropriate normal form and demonstrate use of ACID properties for transaction management.
4. Use advanced database Programming concepts like mongoDB.

Detailed Syllabus:

Unit	Description	Duration [Hrs]
I	**Introduction to Database Management Systems:** Characteristics of Databases, DBMS system architecture, Types of Data Models and Data Modeling, Elements of Database Systems, Entity-Relationship (ER) Model, Entity types: Weak and strong entity sets, Types of Attributes, Keys, Relationship constraints: Cardinality, Extended Entity-Relationship (EER) Model: Generalization, Specialization and Aggregation ,E-R and EER diagrams.	7
II	**SQL and PL/SQL:** Introduction to **SQL:** Data Types and Literals, DDL, DML, DCL, TCL, DQL commands, SQL Operators, Index and Sequence in SQL, Views: Creating, Dropping, Updating using Indexes, Set Operations, Joins, SQL Functions, Nested Queries, **PL/SQL:** Stored Procedures & Functions, Cursors, Triggers, Assertions, Roles and Privileges, Exception handlings	8
III	**Relational Model and Transaction Management:** Introduction to the Relational Model, relational schema and concept of keys, CODD's rules, Pitfalls in Relational-Database designs, Concept of normalization, Function Dependencies, 1NF, 2NF, 3NF, BCNF. **Transaction Management:** Transaction concept, transaction state, ACID properties, serializability, Recoverability: Shadow-Paging and Log-Based Recovery, Checkpoints. Implementation of Isolation, Testing for serializability, Concurrency control, Lock based concurrency control, Time stamping methods.	8

IV	**NoSQL Databases:** Introduction to Distributed Database System- Advantages, disadvantages, CAP Theorem. NoSQL Database: Introduction, need, Features, Types of NoSQL Database, BASE Properties, MongoDB (with syntax and usage): CRUD Operations, Indexing, Aggregation, MapReduce, Replication, Sharding, Unstructured text, Information retrieval systems, document retrieval and ranking	**7**
	Total	**30**

Text Books:

1. Silberschatz A., Korth H., Sudarshan S., "Database System Concepts", McGraw Hill Publishers, 7th Edition, 2020 ISBN 978-0-07-802215-9.
2. Elmasri and Navathe, “Fundamentals of Database Systems”, 5thEdition, Pearson Education
3. Ivan Bayross, "SQL, PL/SQL the Programming Language of Oracle", BPB Publications, 2014 ISBN: 9788176569644.
4. Connally T, Begg C., "Database Systems- A Practical Approach to Design, Implementation and Management", Pearson Education, 5th Edition, 2010, ISBN 81-7808-861-4.
5. Serge Abiteboul, Peter Buneman, Dan Suciu, Data on the Web: From Relations to Semistructured Data and XML / / 1st Edition / ISBN-13: 978-155860622/ ISBN-10: 155860622X

Reference Books:

1. C. J. Date, "An Introduction to Database Systems", Addison-Wesley, 8th Edition, 2004, ISBN 0321189566.
2. S. K. Singh, "Database Systems: Concepts, Design and Application", Pearson Education, 2009,ISBN 9788177585674.
3. Ramakrishnan and Gehrke, Database Management Systems,, McGrawHill 2003
4. R.P. Mahapatra, Database Management Systems,, Khanna Book Publishing 2016.
5. J. D. Ullman, “Principles of Database Systems”, 2nd Ed., Galgotia Publications

e-sources: NPTEL Video Courses:

1. https://nptel.ac.in/courses/106/105/106105175/
2. https://www.mooc-list.com/tags/dbms

Program :	B. Tech. CSE (AI & ML)					Semester:	III
Course :	OE- Mathematics-Applied Mathematics (Suggested for Mechanical branch) (Offered by AS&H)					Code :	BSH23OE01
Credits	Teaching Scheme (Hrs./Week)			Evaluation Scheme and Marks			
	Lecture	Practical	Tutorial	FA		SA	Total
				FA1	FA2		
2	2	-	-	10	10	30	50

Prior knowledge of
1. Univariate Calculus
2. Multivariate Calculus

are essential.

Course Objectives:
After completion of the course, students will have adequate background, conceptual clarity and knowledge of mathematical principles related to:
1. Statistical techniques and Probability theory for Data Analysis.
2. Partial differential equations applied to mechanical engineering problems such as mechanical vibrations and heat transfer.
3. Laplace Transform and Inverse Laplace Transform applied to solve linear differential equations.

Course Outcomes:
After learning the course, the students should be able to:
1. **Apply** descriptive statistical techniques for measures of variability of numerical data, Curve fitting, Correlation and Regression.
2. **Make predictions** for the numerical data using probability theory and hypothesis testing.
3. **Apply** variable separation method to solve wave, transport, one and two-dimensional heat flow equations.
4. **Solve** the mass spring system and similar problems using Laplace and Inverse Laplace Transform.

Detailed Syllabus:

Unit	Description	Duration [Hrs]
I	**Statistics:** Measures of Variability: Standard deviation, Coefficient of variation, Moments, Skewness and Kurtosis, Curve fitting, Correlation and Regression.	7
II	**Probability Distributions:** Probability, Theorems on Probability, Mathematical Expectation, Binomial, Poisson, and Normal Distributions. **Hypothesis Test:** z-test, t-test, Chi-Square test, ANOVA Test.	8
III	**Applications of Partial Differential Equations:** Solution to One dimensional Wave, Heat and Transport equation, Two-dimensional heat flow equation using Method of separation of variables.	7
IV	**Laplace Transform:** Introduction, Laplace Transform of some standard and special functions, Region of convergence and Properties, properties and theorems of Laplace Transformation.Inverse Laplace Transform, Application of Laplace Transform to solve LDE.	8
	Total	30

Text Books:
1. Peter O'Neil, "Advanced Engineering Mathematics", Thomson Learning ,7 Edition, ISBN 13:9781337274524.
2. B.V. Ramana , "Higher Engineering Mathematics", Tata McGraw-Hill, 34 edition, ISBN 13:9780070634190.

Reference Books:

1. Erwin Kreyszig, “Advanced Engineering Mathematics” Wiley Eastern Ltd.,10 Edition, ISBN 13: 9780470458365
2. B. S. Grewal , “Higher Engineering Mathematics”, Khanna Publication, 42 Edition, ISBN 13:9788174091955.
3. N. P. Bali, Manish Goyal, “ A textbook of Engineering Mathematics”, 9th Edition, ISBN 16:978-8131808320
4. H.K. Das, “Advanced Engineering Mathematics”, S Chand & Company Ltd, 22 edition, ISBN 9352533836.

e-sources:

NPTEL Course lectures links:

1. https://nptel.ac.in/courses/111/105/111105038/ (P.D.E)
2. https://onlinecourses.nptel.ac.in/noc20_ma13/ (Advanced Engineering Mathematics)

Program :	B. Tech. CSE (AI & ML)					Semester: III	
Course :	OE- Mathematics-Computational Techniques (Suggested for E&Tc branch) (Offered by AS&H)					Code :	BSH23OE02
Credits	Teaching Scheme (Hrs./Week)			Evaluation Scheme and Marks			
	Lecture	Practical	Tutorial	FA		SA	Total
				FA1	FA2		
2	2	-	-	10	10	30	50

Prior knowledge:

1. Univariate Calculus
2. Multivariate Calculus is essential

is essential

Course Objectives:

After completion of the course, students will have adequate background, conceptual clarity and knowledge of mathematical principles related to:

1. Statistical techniques, Probability theory, and hypothesis techniques.
2. Numerical techniques to approximate solutions for interpolation, integration and ordinary differential equations.

Course Outcomes:

After learning the course, the students should be able to:

1. **Apply** statistical methods like variability, curve fitting, correlation and regression analysis for prediction of a given data.
2. **Analyze** the data using probability theory and distributions.
3. **Make predictions** for the numerical data using hypothesis testing.
4. **Compute** approximate solution for interpolation, integration and ordinary differential equations using numerical methods.

Detailed Syllabus:

Unit	Description	Duration [Hrs]
I	**Statistics:** Measures of Variability: Standard deviation, Coefficient of variation, Moments, Skewness and Kurtosis, Curve fitting, Correlation and Regression.	7
II	**Probability Distributions:** Probability, Theorems on Probability, Random Variable, Probability mass function, Mathematical Expectation, Probability distributions: Binomial, Poisson, and Normal..	8
III	**Sampling Distribution:** Introduction and Types, Population vs Sample **Hypothesis Tests:** Level of significance, Confidence interval, p-Test, z-test, t-test, Chi-Square test, ANOVA Test.	7
IV	**Numerical Methods:** **Interpolation:** Finite Differences, Newton's and Lagrange's interpolation formula. **Numerical Integration:** Trapezoidal and Simpson's rule **Ordinary differential equations:** Euler's, Modified Euler's and Runge-Kutta fourth order methods	8
	Total	30

Text Books:

1. Peter V. Neil, "Advanced Engineering Mathematics", Thomson Learning ,7 Edition, ISBN 13:9781337274524.
2. B.V. Ramana , "Higher Engineering Mathematics", Tata McGraw-Hill, 34 edition, ISBN 13:9780070634190.

Reference Books:

1. M. D. Greenberg , “Advanced Engineering Mathematics”, Pearson Education, 2 Edition, ISBN 13:9780486492797.
2. B. S. Grewal , “Higher Engineering Mathematics”, Khanna Publication, 42 Edition, ISBN 13:9788174091955.
3. N. P. Bali, Manish Goyal, “ A textbook of Engineering Mathematics”, 9th Edition, ISBN 16:978-8131808320
4. H.K. Das, “Advanced Engineering Mathematics”, S Chand & Company Ltd, 22 edition, ISBN 9352533836.

e-sources:

NPTEL Course lectures links:

1. Probability Theory for Data Science - Course (nptel.ac.in) (Statistics, Probability and Hypothesis testing)
2. Numerical methods - Course (nptel.ac.in) (Numerical Methods)

Program :	S.Y.B. Tech.					Semester: III	
Course :	OE- Mathematics-Applied Mathematics (Suggested for Civil branch)					Code :	BSH23OE03
Credits	Teaching Scheme (Hrs./Week)			Evaluation Scheme and Marks			
	Lecture	Practical	Tutorial	FA		SA	Total
				FA1	FA2		
2	2	-	-	10	10	30	50

Prior knowledge:

1. Univariate Calculus
2. Multivariate Calculus is essential

is essential

Course Objectives:

After completion of the course, students will have adequate background, conceptual clarity and knowledge of mathematical principles related to:

1. Statistical techniques, Probability theory, and hypothesis techniques.
2. Higher level mathematics and their applications in E&TC Engineering.

Course Outcomes:

After learning the course, the students should be able to:

1. **Apply** descriptive statistical techniques for measures of variability of numerical data, Curve fitting, Correlation and Regression.
2. **Make predictions** for the numerical data using probability theory and hypothesis testing.
3. **Compute** approximate solution for interpolation, integration and ordinary differential equations using numerical methods.
4. **Examine** the vector fields using concepts of vector differentiation and Integration.

Detailed Syllabus:

Unit	Description	Duration [Hrs]
I	**Statistics:** Measures of Variability: Standard deviation, Coefficient of variation, Moments, Skewness and Kurtosis, Curve fitting, Correlation and Regression.	7
II	**Probability Distributions:** Probability, Theorems on Probability, Mathematical Expectation, Binomial, Poisson and Normal Distribution **Hypothesis Test:** z-test, t-test, Chi-Square test, ANOVA Test.	8
III	**Numerical Methods:** **Interpolation:** Finite Differences, Newton's and Lagrange's interpolation formula **Numerical Integration:** Trapezoidal and Simpson's rule **Ordinary differential equations:** Euler's, Modified Euler's and Runge-Kutta fourth order methods	7
IV	**Vector Differentiation:** Introduction, Vector differential operators, Gradient, Divergent, Curl, Directional Derivatives, Solenoidal and Irrotational fields, Scalar Potential. **Vector Integration and Applications:** Introduction to Line, Surface and Volume Integration of vectors, Work-done, Green's Lemma, Gauss's Divergence theorem, Stoke's theorem, Application to problems in Electro-Magnetic fields.	8
	Total	30

Text Books:

1. B.V. Ramana , "Higher Engineering Mathematics", Tata McGraw-Hill, 34 edition, ISBN 13:9780070634190
2. Peter O'Neil, "Advanced Engineering Mathematics", Thomson Learning ,7 Edition, ISBN 13: 9781337274524

Reference Books:

1. M. D. Greenberg , “Advanced Engineering Mathematics”, Pearson Education, 2 Edition, ISBN 13: 9780486492797
2. S.R.K. Iyengar, Rajendra K. Jain, “Advanced Engineering Mathematics”, Alpha Science International, Ltd,4
3. Edition, ISBN 13: 9781842658468
4. B. S. Grewal , “Higher Engineering Mathematics”, Khanna Publication, 42 Edition, ISBN 13: .9788174091955
5. N. P. Bali, Manish Goyal, “ A textbook of Engineering Mathematics”, 9th Edition, ISBN 16:978-8131808320

e-sources:

NPTEL Course lectures links:

1. **https://nptel.ac.in/courses/111/105/111105090/ (Probability)**
2. **https://onlinecourses.nptel.ac.in/noc20_ma13/ (Advanced Engineering Mathematics)**
3. **V-lab (IIT-Bombay) link: http://vlabs.iitb.ac.in/vlabs-dev/labs/numerical_lab/labs/explist.php**

Program :	**B. Tech. CSE (AI & ML)**	**Semester: IV**	
Course :	**OE- Mathematics-Computational Techniques** (OE Maths Suggested for Comp/IT branch)	**Code :**	**BSH24OE07**

Credits	**Teaching Scheme (Hrs./Week)**			**Evaluation Scheme and Marks**			
	Lecture	**Practical**	**Tutorial**	**FA**		**SA**	**Total**
				FA1	**FA2**		
2	**2**	**-**	**-**	**10**	**10**	**30**	**50**

Prior knowledge of
1. Univariate Calculus
2. Multivariate Calculus

is essential.

Course Objectives:
After completion of the course, students will have adequate background, conceptual clarity and knowledge of mathematical principles related to:
1. Statistical techniques, Probability theory, and hypothesis techniques.
2. Different mathematical approaches for optimization.

Course Outcomes:
After learning the course, the students should be able to:
1. **Apply** statistical methods like variability, curve fitting, correlation and regression analysis for prediction of a given data.
2. **Analyze** the data using probability theory and distributions.
3. **Make predictions** for the numerical data using hypothesis testing.
4. **Solve** transportation and assignment problems using optimization techniques..

Detailed Syllabus:

Unit	**Description**	**Duration [Hrs]**
I	**Statistics:** Measures of Variability: Standard deviation, Coefficient of variation, Moments, Skewness and Kurtosis, Curve fitting, Correlation and Regression.	**7**
II	**Probability Distributions:** Probability, Theorems on Probability, Random Variable, Probability mass function, Mathematical Expectation, Probability distributions: Binomial, Poisson, and Normal.	**8**
III	**Sampling Distribution:** Introduction and Types, Population vs Sample **Hypothesis Tests:** Level of significance, Confidence interval, p-Test, z-test, t-test, Chi-Square test, ANOVA Test.	**7**
IV	**Transportation Problems:** Introduction, Mathematical model of transportation problem, Methods of finding initial solutions: North-west Corner rule, Least cost method, VOGEL"s approximation method, Optimality of initial solution using MODI Method. **Assignment Problems:** Introduction, Mathematical model of Assignment problem, solutions to Assignment problems using Hungarian method.	**8**
	Total	**30**

Text Books:
1. Peter O'Neil, "Advanced Engineering Mathematics", Thomson Learning ,7 Edition, ISBN 13:9781337274524.
2. B.V. Ramana , "Higher Engineering Mathematics", Tata McGraw-Hill, 34 edition, ISBN 13:9780070634190.

Reference Books:

1. M. D. Greenberg , "Advanced Engineering Mathematics", Pearson Education, 2 Edition, ISBN 13:9780486492797.
2. B. S. Grewal , "Higher Engineering Mathematics", Khanna Publication, 42 Edition, ISBN 13:9788174091955.
3. N. P. Bali, Manish Goyal, " A textbook of Engineering Mathematics", 9th Edition, ISBN 16:978-8131808320
4. H.K. Das, "Advanced Engineering Mathematics", S Chand & Company Ltd, 22 edition, ISBN 9352533836.

e-sources:

NPTEL Course lectures links:

1. Probability Theory for Data Science - Course (nptel.ac.in) (Statistics, Probability and Hypothesis testing)
2. https://nptel.ac.in/courses/110/106/110106059/(Transportation&AssignmentsProblems)

Program :	B. Tech. CSE (AI & ML).	Semester: IV	
Course :	OE- Mathematics-Mathematical Optimization (Suggested for AI & ML branch)	Code :	BSH24OE08

Credits	Teaching Scheme (Hrs./Week)			Evaluation Scheme and Marks			
	Lecture	Practical	Tutorial	FA		SA	Total
				FA1	FA2		
2	2	-	-	10	10	30	50

Prior knowledge of

1. Univariate Calculus
2. Multivariate Calculus

is essential.

Course Objectives:

After completion of the course, students will have adequate background, conceptual clarity and knowledge of mathematical principles related to:

1. Different mathematical approaches for optimization.
2. Commonly used tools and techniques in network analysis.

Course Outcomes:

After learning the course, the students should be able to:

1. **Formulate** and solve linear programming models using graphical, Simplex method.
2. **Solve** transportation and assignment problems using optimization techniques.
3. **Analyze** the project network problems and their solutions using critical path method to optimize models.
4. **Apply** variants of numerical methods to find optimal solutions for constrained, unconstrained problems.

Detailed Syllabus:

Unit	Description	Duration [Hrs]
I	**Linear Programming (LP):** Introduction, formulation of Linear Programming problems, Graphical solution method, multiple optimal solutions, Unbounded solutions, Infeasible solutions, Simplex Method.	8
II	**Transportation Problems:** Introduction, Mathematical model of transportation problem, methods of finding initial solutions: North-west Corner rule, Least cost method, VOGEL"s approximation method, Optimality of initial solution using MODI Method. **Assignment Problems:** Introduction, Mathematical model of Assignment problem, solutions to Assignment problems using Hungarian method.	7
III	**Network Analysis:** Network Diagram, Project Management: PERT and CPM, Critical path analysis, Project scheduling with uncertain activity time, and Project time-cost	8
IV	**Unconstrained optimization:** One-dimensional search methods, Gradient-based methods, Conjugate direction and quasi-Newton methods, **Constrained Optimization:** Lagrange theorem	7
	Total	30

Text Books:

1. Rao S. S., Engineering Optimization Theory and Practice, Willy Eastern Ltd.4th Edition, ISBN: 978-0-470- 18352-6
2. Taha Hamdy, Operation Research: An Introduction, Pearson Education,9th Edition, ISBN:0134444019

Reference Books:

1. Sharma S.D.Operation Research, Kedarnath Ramnath & Co.Edition, ISBN: 9380803389
2. Peter. O'Neil, "Advanced Engineering Mathematics," Thomson Learning,7 Edition, ISBN 13: 9781337274524
3. Hira Gupta, "Operation Research," S.Chand Publication, ISBN(13): 9788121909686.
4. Sharma J.K."Operations Research-Theory and Applications," Trinity Press,6 Edition, ISBN:9789385935145

e-sources:

NPTELCourselectureslinks:

1. https://nptel.ac.in/courses/111/102/111102012/(LPP)
   https://nptel.ac.in/courses/110/106/110106059/(Transportation&AssignmentsProblems)

Program :	B. Tech CSE (AI & ML)					Semester:	IV
Course :	EEM-II: Designing Thinking & Innovation Management (Offered by Department of Applied Sciences & Humanities)					Code :	BSH24EM02
Credits	Teaching Scheme (Hrs./Week)			Evaluation Scheme and Marks			
	Lecture	Practical	Tutorial	FA		SA	Total
				FA1	FA2		
2	2	-	-	10	10	30	50

Prior knowledge : NIL

Course Objectives: This course aims at enabling students :

1. To introduce the students to the concept of Design Thinking and its relevance in innovation
2. To equip students with the core concepts, frameworks, and techniques of Innovation management and its Applications.
3. To help students to understand design thinking as a creative problem-solving approach

Course Outcomes:

After learning the course, the students will be able to:

1. Explain the concept **o**f Design Thinking and Opportunity Assessment in Business
2. Demonstrate strategic foresight for the business models
3. Apply the concept of Innovation Management in for business growth.
4. Apply the techniques of Internet Business Design for business growth

Detailed Syllabus:

Unit	Description	Duration [Hrs]
I	**Introduction to Design Thinking:** Meaning of Design Thinking, Design thinking for competitive advantage, The Need for Creative and Design, why design needs entrepreneurial mindset, combining entrepreneurial and design thinking, Opportunity Assessment - "How do we source and identify opportunities?" and "Which opportunities should we invest time and money in?" Three dimensions of Opportunity Assessment - Product-Market fit, Product-Company fit, and Product-Business fit. "Go/No-Go" recommendation to invest and initiate.	7
II	**Business Challenges and Design Thinking Solutions Paradigm Shift:** The seven steps of design thinking, Tools of Design Thinking, First Mover and late mover Advantage, Five Cs of Opportunity Storytelling, Strategic Foresight, Sensing, Value Redefinition, Experience Design, Humanization, Prototyping, Business Model Design	8
III	**Innovation Management:** Concept of Innovation, Characteristics of innovation, Sources of innovation, Types of innovation, Levels of Innovation, Evolution of innovation management, Organizations and Innovation Process, Diffusion of Innovation, Effective innovation management, Performance evaluation, Risk Management in Innovation. Case Studies based on Innovation management	7
IV	**Internet Business Design:** Digital/Internet Business Model, Design of Services and Customer Experience. Service sector – IDEO, Lego, E-commerce market players design thinking strategies. Design Thinking and IoT Toward Sustainable Design Thinking. Managing Future Technologies, and minimizing risk of failure.	8
	Total	30

Text Books:

1. Robert Curedale, Design Thinking Process and Methods,5th Edition.

Reference Books:

1. Walter Brenner, Falk Uebernickel, Design Thinking for Innovation, Springer Link, 2016.
2. Christian Müller-Roterberg, Handbook of Design Thinking, Kindle Direct Publishing,ISBN: 978-1790435371
3. Anuja Agarwal, Design Thinking: A framework for applying Design Thinking in Problem Solving, CL India

e-sources:

1. Roger Martin, "The Design of Business: Why Design Thinking is the Next Competitive Advantage",Harvard Business Press , 2009.
2. https://www.designdisciplin.com/the-story-of-design-thinking/
3. https://online.hbs.edu/blog/post/what-is-design-thinking

Program :	B. Tech CSE (AI & ML)					Semester:	IV
Course :	EEM-II: Project Management (Offered by Department of Applied Sciences & Humanities)					Code :	BSH24EM03
Credits	Teaching Scheme (Hrs./Week)			Evaluation Scheme and Marks			
	Lecture	Practical	Tutorial	FA		SA	Total
				FA1	FA2		
2	2	-	-	10	10	30	50

Prior knowledge : NIL

Course Objectives:

This course aims at enabling students :

1. To help the students gain understanding regarding the concept of projects and Project Management
2. To enable the students to know the key components of project management including time, cost & Risk.
3. Recognize issues in a realistic project scenario to understand design thinking as a creative problem-solving approach

Course Outcomes:

After learning the course, the students will be able to:

1. Explain different stages of project and their significance.
2. Demonstrate the triple constraints concepts in project management.
3. Apply appropriate approaches to plan execute and evaluate projects through case studies.
4. Analyze to mitigate the risk associated with projects.

Detailed Syllabus:

Unit	Description	Duration [Hrs]
I	**Introduction to Project Management :** Concept and Definition of Project, Characteristics of Project, Concept and definition of Project Management, Functions of Project Management, Importance of Project Management, Who is a Project Manager, Roles & Responsibilities of Project Manager. Understanding the Phases in the Lifecycle of Projects and their Significance, Different types of Projects: Industrial, Telecommunication, Research and more, Project Selection Methods : Agile method , Waterfall, Methods, Scrum Model & Kanban Model, Jira Model	7
II	**The Triple Constraint in Project Management :** The concept of the Triple Constraint in Project Management : Scope, Cost and Time, Project Cost Management : Concept, Consideration, Five types of Costs involved in a project, Cost Management process, Project Time Management and methods of Time estimation, Communications Management in Project , Work Breakdown Structure (WBS). Case studies based on Mega Projects of the World.	7

III	**Planning and Execution of Project:** Developing a Mission, Vision, Goals of the project. Concept and definition of Project Planning. Importance of Project Planning. Concept of Project Execution, Phases of Project Execution, Project Evaluation; The Review Technique – Planning and Scheduling of Activity, Networks - Concept of PERT/CPM, Assumptions in PERT Modeling – Time-cost, Trade-offs, HRM issues in Project Management & How they can be tackled, Reasons for Failures of Project , Case Study with respect to different Domains	**8**
IV	**Project Monitoring and Risk Management :** Concept of Project Monitoring , How to Building a Suitable Monitoring; Control System, Concept of Conflict Management, Concept & Definition of Risk and Risk Management, Concept of Risk Matrix Analysis, Strategies to Manage Risks, An Overview of Useful Techniques and Tools Used in Project Management. Case Studies with respect to different Domains.	**8**
	Total	**30**

Text Books:

1. Joseph Heagney, Fundamentals of Project Management, American Management Association, 2012

Reference Books:

1. Erik W Larson, Clifford Gray, Rohit Joshi; Project Management-The managerial process, MacGraw Hill Publication, 2021
2. Punmia, Project Management with CPM /PERT, Laxmi Publications, 2001
3. Robert L Kimmons, Project Management Basics, Taylor & Francis Ltd, 2018
4. N. D. Vohra, Quantitative Techniques in Management, Tata McGraw Hill Book Co. Ltd.

e-sources:

1. https://www.entrepreneur.com/
2. http://dst.gov.in/scientific-programme/t-d-tdb.htm
3. https://www.youtube.com/

Program :	B. Tech CSE (AI & ML)					Semester:	IV
Course :	EEM-II: Fostering Entrepreneurship and Startups (Offered by Department of Applied Sciences & Humanities)					Code :	BSH24EM04
Credits	Teaching Scheme (Hrs./Week)			Evaluation Scheme and Marks			
	Lecture	Practical	Tutorial	FA		SA	Total
				FA1	FA2		
2	2	-	-	10	10	30	50

Prior knowledge : NIL

Course Objectives: This course aims at enabling students :
1. To inculcate an entrepreneurial mindset into the minds of young professionals
2. To enable the students to evaluate challenges relating to new ventures.
3. To provide the students an understanding about different skills for founding, leading & managing startups .

Course Outcomes:
After learning the course, the students will be able to:
1. Explain the concepts and role of entrepreneurship.
2. Identify new venture opportunities in today's Business world.
3. Analyze strategic choices in relation to the growth of new ventures
4. Correlate the Startup opportunities and key factors supporting the Start up Culture in India.

Detailed Syllabus:

Unit	Description	Duration [Hrs]
I	**The Entrepreneurial Perspective :** Why to become entrepreneur ,Characteristics of an Entrepreneur, Functions of an Entrepreneur, Types of Entrepreneurs , Distinction between Entrepreneur and Manager The entrepreneurial decision process, Role of entrepreneur in economic development, future of entrepreneurs, Concept of Business Opportunity, Business Opportunities Identification Process..	7
II	**Creating & Starting Ventures :** Concept of business idea, Sources of new idea generation, Methods of Idea generation, Creative problem solving, Product planning and development, Business Structure, Creating a Business Plan, Market Size Analysis, Legal issues and Regulations to set up a business, Business Patents, Trademarks, Copy rights, Trade secrets, Licensing.	7
III	**Managing and Growing a New Business Venture :** Attractiveness of a new entry opportunity- Entry strategy for new entry exploitation- risk reduction strategies for new entry exploitation, Growth Strategies – economic implications of growth, overcoming pressures on existing financial resources, human resources, management of employees and entrepreneur's time, Sustaining a Business turbulence, Government incentives & Schemes.	8

IV	**The Startup Ecosystem in India :** Meaning of Startup, Types of Start-ups, The Rise of The Startup Economy, Startup Policy, Startup opportunities, and Financial support for Start ups. Recent initiatives including Start up India, Make in India, Digital India, and Policies for technology Start-ups,E- commerce Startups, Tech Support and Proto type Development centers. Start up Infrastructures: Co – Working Space, Market development initiatives..	**8**
	Total	**30**

Text Books:

1. C. B. Gupta and N. P. Srinivasan, Entrepreneurial Development, Sultan Chand & Sons, New Delhi, 2008

Reference Books:

1. Kathleen R Allen, Launching New Ventures, An Entrepreneurial Approach, Cengage Learning, 2016.
2. Peter F. Drucker, Innovation and Entrepreneurship.
3. Satish Taneja, S.L.Gupta, Entrepreneurship Development New Venture Creation
4. Donald F Kuratko, Jeffrey S. Hornsby, New Venture Management: The Entrepreneur's Road Map, 2e, Routledge
5. Vasant Desai ,Dynamics of Entrepreneurship Development

e-sources:

1. https://www.entrepreneur.com/
2. http://dst.gov.in/scientific-programme/t-d-tdb.htm
3. https://www.youtube.com/

Program :	B. Tech CSE (AI & ML)					Semester:	IV
Course :	EEM-II: Business Finance for Engineers (Offered by Department of Applied Sciences & Humanities)					Code :	BSH24EM05
Credits	Teaching Scheme (Hrs./Week)			Evaluation Scheme and Marks			
	Lecture	Practical	Tutorial	FA		SA	Total
				FA1	FA2		
2	2	-	-	10	10	30	50

Prior knowledge : NIL

Course Objectives:

This course aims at enabling students

1. The inculcate the knowledge of the core concepts of business finance and its importance in managing a business
2. To improve students 'understanding of the time value of money concept and the role of finance in the current competitive business scenario..

Course Outcomes:

After learning the course, the students will be able to :

1. Explain the concept of Business Finance and financial planning.
2. Illustrate the concept of capitalization in Business Organizations.
3. Analyze Financial markets and the role of financial institutions in Business Development.
4. Evaluate the role of Financial System in Business.

Detailed Syllabus:

Unit	Description	Duration [Hrs]
I	**Business Finance:** Concept of Business Finance, Objective and Scope, Significance of Finance, principles of business finance, Finance Function, Business finance v/s Corporate finance, Source of Finance. Meaning of financial planning, steps in financial planning, significance of financial planning, essential features of a good financial plan, Personal financial planning, Types of financial plan.	7
II	**Capitalization :** Introduction to capitalization, Amount of Capitalization, Over Capitalization, Under Capitalization, Venture Capital - Meaning of Venture Capital, Method of Venture Financing ,Venture Capital Funds, Polices and Procedures adopted by Venture Capitalists, Venture Capital in India, Guidelines for Venture Funds.	7
III	**Financial Markets, Institutions and Instruments:** Introductions to Financial Markets: Nature, Functions and Types of Financial markets, Different Financial Instruments, Sources of financing: Shares, Debentures, Term Loans, Retained Earnings, Public Deposits, Bonds, Trade Credit, Mutual Funds - Definition, Types of Mutual Funds, Significance of Mutual Funds; Case Studies on Financial Markets.	8

IV	**Constituents of the Financial System and Regulatory Institutions :** Classification: Fund Based, Non Fund Based and Modern Services, Hire Purchasing, Leasing: Lease Financing - Essential Elements of Leasing, Types of Leases, Merits and Demerits of Lease Financing, Merchant Banking - Role; Functions of Merchant Banking, Factoring. Debt management, Portfolio Management. RBI - Organization, objectives, role and functions, monetary policy of RBI, NABARD, SEBI - Organization and Objectives	**8**
	Total	**30**

Text Books:

1.Srivastava, R.M. Essentials of Business Finance, Himalaya Publishing House, Kalyani Publication**s**

Reference Books:

1. Gordon, E. & Natarajan, K. Financial Markets and Institutions, Himalaya Publishing House.
2. Khan and Jain, Financial Management, Tata McGraw Hill, 2008
   Singh, Preeti. Investment Management. Himalaya Publishing House,
3. Kale, N.G. Business Organization. Manisha Publications.

e-sources:

1. https://www.youtube.com/watch?v=TgF2XvjquUU&list=PLLy_2iUCG87CXY2B6fPex1SOIqxzzD5Wj
2. https://www.youtube.com/watch?v=CCQwz_Gwo6o
3. https://www.youtube.com/watch?v=OT5RdoJAkhY&list=PLPjSqITyvDeUTeAOGhip_ubjN3y8oqT13

Program :	B. Tech. CSE (AI & ML)					Semester:	IV
Course :	Professional Development Training					Code :	BSH24AE05
Credits	Teaching Scheme (Hrs./Week)			Evaluation Scheme and Marks			
	Lecture	Practical	Tutorial	TW	OR	PR	Total
2	-	4	-	100	-	-	100

Prior knowledge: Basic Mathematics & English

Course Objectives:

This course aims at enabling the students

1. To enhance the logical reasoning skills of the students and improve the problem-solving abilities.
2. To improve the overall professional development of students.

Course Outcomes:

After learning the course, the students will be able to:

1. **Apply** mathematical concepts to solve diverse numerical problems encountered in engineering, spanning arithmetic, algebra, geometry, and statistics.
2. **Employ** deductive reasoning, interpret data, and discern patterns to navigate complex logical puzzles and analytical challenges typical of engineering aptitude tests and professional scenarios.
3. **Identify** grammatical nuances, enhancing their verbal and written communication prowess essential for effective engineering discourse.

Detailed Syllabus:

Unit	Description	Duration [Hrs]
I	**Numerical Ability-I** HCF & LCM and Number System, Geometry, Ages, Allegations and Mixtures, Averages, Clocks and Calendars, Equations, Percentages,Profit and Loss, Ratios and Proportion, Series and Progressions, Time, Speed and Distance, Time and Work.	16
II	**Numerical Ability-II** Permutations and Combinations, Probability, Mean, Median, Mode, Standard Deviation, and Variance, Data Interpretation, Graphical Data Interpretation, Pie Charts, Tabular Data Interpretation, Simple Arithmetic Operations, Interest and Compound Interest, Linear equations, Quadratic equations, Triplets, Trigonometry	16
III	**Logical Reasoning** Clocks and Calendar, Direction sense, Family tree, Syllogism, Seating arrangement, Team formation, Coding and Decoding, Number Series and Letter Series, Ranking and Arrangements, Game-Based Aptitude.	8
IV	**Verbal Ability & Reading Comprehension** Subject-Verb Agreement, Articles and Other Determiners, Prepositions, Tenses, Parts of Speech, Active and Passive Voice, Direct and Indirect Speech, Conjunctions, Sentence Selection, Contextual Vocabulary, Error Spotting and Sentence Correction, Sentence Completion, Synonyms and Antonyms, Reading Comprehension, Jumble words & sentences.	10
	Total	50

Text Books: **1.** Arun Sharma, Quantitative Aptitude, 2016, 7th Edition, McGraw Hill Education Pvt. Ltd. 2. ETHNUS, Aptimithra, 2013, 1stEdition, McGraw-Hill Education Pvt.Ltd.
Reference Books: **1.** R S Aggarwal, Quantitative Aptitude For Competitive Examinations, 2017, 3rd Edition, S. Chand Publishing, Delhi. 2. M. Tyra, Quicker Maths, 2018, 5th edition, 2018, BSC publishing company Pvt. Lt.

Program:	B. Tech. CSE (AI & ML)					Semester:	IV
Course:	Database Management System Lab					Code:	BCS24PC08
Credits	Teaching Scheme (Hrs./Week)			Evaluation Scheme and Marks			
	Theory	Practical	Tutorial	TW	PR	OR	Total
02	-	04	-	50	50		100

Prior Knowledge of : Data Structures is essential.

Course Objectives:

1. To develop Database programming skills
2. To develop basic Database administration skills.
3. To learn different types of database systems and their applications in different scenarios.
4. To develop skills to handle SQL database

Course Outcomes:

After learning the course, students will be able to:

1. Implement SQL queries for given requirements.
2. Implement PL/SQL Code block for given requirements
3. Apply practical principles involved in the design and use of database systems.
4. Design and implement databases for various scenarios.

Guidelines for Students:

Continuous assessment of laboratory work is based on overall performance and Laboratory assignments performance of students.

Guidelines for Laboratory Conduction:

The instructor is expected to frame the assignments by understanding the prerequisites, technological aspects, utility and recent trends related to the topic. The assignment framing policy needs to address the average students and is inclusive of an element to attract and promote the intelligent students. Use of open source software is encouraged. Based on the concepts learned. Instructors may also set one assignment or mini-project that is suitable to each branch beyond the scope of the syllabus. Operating System recommended :- 64-bit Open source Linux or its derivative Programming tools recommended: - MYSQL/Oracle, MongoDB, ERD plus, ER Win

Assignment No.	Suggested List of Assignments
1	Design at least 10 queries to perform SQL DDL and DML operations.
2	Design at least 10 queries to perform DML operations , use appropriate constraints (Primary key, foreign key, default, check etc.)
3	Design and Develop SQL DDL statements which demonstrate the use of SQL objects such as Table, View, Index, Sequence, Synonym
4	Design at least 10 SQL queries for suitable database application using SQL DML statements: Insert, Select, Update, Delete with operators, functions, and set operators.
5	Design at least 10 SQL queries for suitable database application using SQL DML statements: all types of Join, Sub-Queries.

6	Write a PL/SQL block of code. Instructor can frame 5 problem statements for writing PL/SQL block
7	Write a PL/SQL block of code to implement all types of cursors.
8	Write a PL/SQL Stored Procedure and functions. Instructor can frame 5 problem statements.
9	Write a database trigger on the Employee table. Instructors can frame the problem statements for writing Database Triggers of all types.
10	Large Scale Databases: a. Design and Develop MongoDB Queries using CRUD operations. b. Implement aggregation and indexing with suitable examples using MongoDB. c. Implement Map reduce operations with suitable examples using MongoDB. d. Create simple objects and array objects using JSON. Encode and Decode JSON Objects using Java/Perl/PHP/Python/Ruby
11	Implement document retrieval and ranking using any algorithm.
12	Mini Project: a. Write a program to implement MongoDB database connectivity with PHP/ python/Java Implement Database navigation operations (add, delete, edit etc.) using ODBC/JDBC. b. Implement MYSQL/Oracle database connectivity with PHP/ python/Java Implement Database navigation operations (add, delete, edit,) using ODBC/JDBC. c. Students in the group are expected to design and develop a database application with following details: ● Design Entity Relationship Model, Relational Model, Database Normalization ● Front End: Java/Perl/PHP/Python/Ruby/.net ● Backend: MongoDB/MYSQL/Oracle ● Database Connectivity: ODBC/JDBC ● Project Report preparation

Reference Books:

1. Ivan Bayross, “SQL, PL/SQL: The Programming Language of Oracle”, BPB Publication, ISBN10: 8176560723; ISBN-13: 78-8176560726
2. Kristina Chodorow, Michael Dirolf, “MangoDB: The Definitive Guide”, O’Reilly Publications, ISBN: 9781449381561
3. Import, Tidy, Transform,” R for Data Science”, O'REILLY, ISBN: 13:978-93-5213-497-7
4. http://www.tutorialspoint.com/json/ & http://docs.mongodb.org/manual/

Program:	B. Tech. CSE (AI & ML)					Semester:	IV
Course:	Advanced Data Structures Lab					Code:	BCS24PC09
Credits	Teaching Scheme (Hrs./Week)			Evaluation Scheme and Marks			
	Theory	Practical	Tutorial	TW	PR	OR	Total
01	-	02	-	25	25		50

Prior Knowledge of Data structures is essential.

Course Objectives:

1. Understand the theoretical foundations of advanced data structures.
2. Implement advanced data structures efficiently in code.
3. Understand basic techniques and strategies of algorithm analysis.

Course Outcomes:

After learning the course, students will be able to:

1. Choose most appropriate data structures and apply algorithms for graphical solutions of the given problems.
2. Apply and analyze algorithm design techniques using multi way searching and file organization to solve real-world problems effectively.
3. Evaluate trade-offs between different Tree data structures based on problem requirements.
4. Demonstrate proficiency in designing and optimizing Hash functions for specific applications.

Guidelines for Laboratory Conduction:

1. The instructor is expected to conduct a minimum 6 assignments from the suggested list.
2. The instructor is expected to address the theory concepts related to each assignment.
3. Encourage students for appropriate use of Hungarian notation, proper indentation and comments.
4. Use of open source software is to be encouraged.
5. Assignments should be implemented in C++ language.
6. Operating System recommended: - 64-bit Open source Linux or its derivative.
7. Set of suggested assignments list is provided in groups - A and B. Each student must perform at least 6 assignments. (At least 3 from each Group)

Programming tools recommended: - G++/GCC, Eclipse, QT creator

Suggested List of Assignments

Assignment No.	Group A
1	Suppose you are given the following set of keys to insert into a hash table that holds exactly 11 values: 113 , 117 , 97 , 100 , 114 , 108 , 116 , 105, 99. Implement open addressing & chaining strategy to insert all keys into Hash Table
2	A hospital is looking to optimize its bed management system to ensure efficient utilization of resources. They have multiple wards, each with a varying number of beds. Patients arrive at the hospital at different times, requiring different levels of care. The hospital wants to maximize the utilization of beds while ensuring that patients are assigned to appropriate wards based on their medical condition and the availability of beds. Design a solution using appropriate data structure to maintain the priority of patients waiting for beds and efficiently manage the allocation and reallocation of beds within the hospital wards.

3	Given a sorted array keys[0.. n-1] of search keys and an array freq[0.. n-1] of frequency counts, where freq[i] is the number of searches to keys[i]. Implement a binary search tree of all keys such that the total cost of all the searches is as small as possible.
4	A Dictionary stores keywords & its meanings. Provide facility for adding new keywords, deleting keywords, updating values of any entry. Provide a facility to display whole data sorted in ascending/ Descending order. Also find how many maximum comparisons may require for finding any keyword. Use Height balanced tree and find the complexity for finding a keyword
	Group B
5	Develop a Library Management System for a university library using C++. The system needs to efficiently manage the books in the library, including operations such as adding new books, removing books, and searching for books based on various criteria. To optimize the performance of these operations, you decide to implement a B-tree data structure to organize the books in the library. Insertion Operation: ● Develop a function to add new books to the library while maintaining the sorted order of the books in the B-tree. ● When inserting a new book, ensure that the B-tree structure is updated appropriately to accommodate the new book. Deletion Operation: ● Implement a function to remove books from the library while ensuring that the B-tree structure remains balanced. ● When deleting a book, make sure to update the B-tree structure to reflect the removal of the book from the library.
6	You are tasked with developing a Student Records Management System for a university using C++. The system needs to efficiently manage a large dataset of student records, including information such as student ID, name, age, GPA, and major. To ensure fast retrieval of student records based on different criteria, you decide to implement a B+ tree index on the student ID attribute. Insertion Operation: ● Develop a function to insert new student records into the database efficiently. ● When inserting a new student record, ensure that the B+ tree index is updated accordingly to include the new student ID. Deletion Operation: ● Implement a function to delete student records from the database while ensuring that the B+ tree index remains consistent. ● When deleting a student record, make sure to update the B+ tree index to remove the corresponding student ID.

7	Develop a banking application using C++ to manage customer account information for a financial institution. The application needs to store and retrieve customer account details efficiently, allowing quick access to account information for various banking operations. To achieve this, you decide to implement a direct access file. Insertion Operation: ● Develop a function to add new customer accounts to the direct access file. ● When inserting a new account, ensure that it is stored in the appropriate location within the file based on the account number. Update Operation: ● Implement a function to update existing customer accounts in the direct access file. ● Allow users to modify account information such as balance, account type, and customer details. Search Operation: ● Develop search functionality to retrieve customer accounts based on their account numbers. ● Utilize the direct access file structure to perform fast retrieval of account details with minimal time complexity.
8	Company maintains employee information such as employee ID, name, designation and salary. Allow users to add, delete information about employees. Display information of a particular employee. If an employee does not exist an appropriate message is displayed. If it is, then the system displays the employee details. Use index sequential files to maintain the data.

Text Books:

1. Horowitz,Sahani,Dinesh Mehata,"FundamentalsofDataStructuresinC++",Galgotia Publisher, ISBN: 8175152788, 9788175152786.
2. MFolk,BZoellick,G.Riccardi,"FileStructures",PearsonEducation", ISBN:81-7758-37-5
3. PeterBrass,"AdvancedDataStructures",CambridgeUniversityPress,ISBN:978-1-107- 43982-5

Reference Books:

1. A. Aho, J.Hopcroft, J. Ulman,"DataStructures and Algorithms", Pearson Education, 1998,ISBN-0-201-43578-0.
2. Michael J Folk, "File Structures an Object Oriented Approach with C++", Pearson Education, ISBN: 81-7758-373-5.
3. SartajSahani,"DataStructures,AlgorithmsandApplicationsinC++",SecondEdition, University Press, ISBN: 81-7371522 X.
4. GAVPai,"DataStructuresandAlgorithms",McGraw-HillCompanies,ISBN- 9780070667266.
5. Goodrich,Tamassia,Goldwasser,"DataStructuresandAlgorithmsinJava",Wiley Publication, ISBN: 9788126551903

e-sources:

1. https://nptel.ac.in/courses/106102064
2. https://nptel.ac.in/courses/106105085
3. https://nptel.ac.in/courses/106106127

Program:	B. Tech. CSE (AI & ML)				Semester:	IV	
Course:	Full Stack Development - I				Code:	BCS24VS03	
Credits	Teaching Scheme (Hrs./Week)			Evaluation Scheme and Marks			
	Theory	Practical	Tutorial	TW	PR	OR	Total
02	-	04	-	50	50	–	100

Prior Knowledge of Basic Programming is essential.

Course Objectives:

1. Understand the fundamentals of web development including HTML, CSS, and JavaScript.
2. Gain proficiency in back-end development using languages like Node.js, Java & Python.
3. Work on real-world projects to apply theoretical knowledge and gain practical experience in full stack development.

Course Outcomes:

After learning the course, students will be able to:

1. Apply JavaScript concepts such as arrays, loops, iterators, and objects to build interactive functionality.
2. Implement CRUD operations for managing tasks in an online task manager application.
3. Develop a backend application including features such as authorization/authentication, file system operations, HTTP server setup, database interaction, and REST API integration.
4. Implement features such as user authentication, data persistence, and interaction with external APIs to create a fully functional web application.

Guidelines for Laboratory Conduction:

1. The instructor is expected to conduct all the assignments from the suggested list.
2. Encourage students for appropriate use of Hungarian notation, proper indentation and comments.
3. Use of open source software is to be encouraged.
4. Assignments should be implemented as per required language.
5. Operating System recommended: - 64-bit Open source Linux or its derivative.

Programming tools recommended: - G++/GCC, Browser,Text Editors

Assignment No.	Suggested List of Assignments
1	**Build a landing page using HTML Concepts:** Basic tags, HTML standards, HTML tables, HTML forms, semantic HTML.
2	Design and develop a responsive portfolio website using HTML and CSS Concepts: Setup and syntax, Selectors, Visual Rules, Box model, HTML Flow, Color, Typography.
3	Responsive Layout Design Using CSS Flexbox, Grid, and Advanced Grid Techniques Concepts: Flexbox, Grid, Advance grid
4	Create a button using HTML and apply Cascading Style Sheets (CSS) and use javascript to show an alert on the button click displaying the message button clicked concepts: Introduction: console, comments, datatype(built ins), arithematic operators, string concatenation, properties, methods, built-in objects, variables, conditional Conditional Statements, functions, Scope.

5	Build a calculator app using HTML CSS AND javascript concepts: Arrays, Loops, Iterators, objects, Advanced objects.
6	Develop DOM-based Interactive Memory Game concepts: Introduction to DOM, document object, getElementById, getElementByTagName and className, querySelector and querySelectorAll, innerHtml, textContent, innerText, manupulating attrs, changing styles, classList, traversing parent, child, sibling, append & appendChild, removeChild. Introduction to events: Window Object
7	Develop Weather Forecast Application using OpenWeatherMap API Contents: Classes, constructor, instance, methods, methods call, Inheritance, Static methods Async JS, Call stack, WebAPIs on single thread, Callback Hell, Promise, Promise object, setTimeout(), Consuming Promise, Success , failure Callback, catch(), chaining multi Promise, avoiding common mistakes, Promise.all() Async keyword and function, await operator, dependent Promise, handling errors, handling independent Promise, await Promise.all(), getRequest(), postRequest(), Errors and Error handling.
8	Node JS Setup es6 javascript and how to install package Concept: Introduction of Modules in ES6
9	Develop Online Task Manager Using Node.js Concepts: Node REPL, Authorization and Authentication, file system, operating system modules, Events module, npm packages and understanding of dependencies, package.json, Setting up server with HTTP, Interacting with a database, Interacting with Backend API, Introduction to REST API.
10	responsive project.

Text Books:

1. Jon Duckett, “HTML and CSS: Design and Build Websites”, 1st Edition, ©2011 by John Wiley & Sons, Inc., Indianapolis, Indiana ISBN: 978-1-118-00818-8.
2. Andy Budd, Simon Collison, and Cameron Moll, “CSS Mastery: Advanced Web Standards Solutions”, 3rd Edition, 2016.
3. David Choi and Nathanial Schutta, “Full-Stack React, TypeScript, and Node: Build scalable, modular web applications with TypeScript, React, and Node.js”, 1st Edition, 2020

Reference Books:

1. Jon Duckett, “JavaScript and JQuery: Interactive Front-End Web Development”, 1st Edition, 2016.
2. Marijn Haverbeke, “Eloquent JavaScript: A Modern Introduction to Programming”, 3rd Edition, 2018.
3. Mario Casciaro, “Node.js Design Patterns”, 2nd Edition, 2016.
4. David Herron, “ Node.js Web Development”, 1st Edition, 2016

e-sources:

1. https://www.w3schools.com/html/
2. https://developer.mozilla.org/en-US/docs/Web/HTML
3. https://css-tricks.com/snippets/css/complete-guide-grid/
4. https://developer.mozilla.org/en-US/docs/Web/JavaScript
5. https://www.tutorialspoint.com/nodejs/index.htm
6. https://developer.mozilla.org/en-US/docs/Web/API/Fetch_API
7. https://www.javascripttutorial.net/dom/
8. https://openweathermap.org/apiha

Program:	B. Tech. CSE (AI & ML)					Semester:	IV	
Course :	Constitution of India					Code:	BSH24VE02	
Credit	Teaching Scheme			Evaluation Scheme				
	Lecture	Practical	Tutorial	FA		TW	SA	Total
				FA1	FA2			
2	2	-	-	20	30	-	-	50

Course Objectives:

1. Provide an overview of the historical context leading to the framing of the Indian Constitution.
2. Study the fundamental rights enshrined in the Constitution and their application.
3. Study the structure and functions of key constitutional institutions like the President, Parliament, and Judiciary.

Course Outcomes:

After learning the course, the students will be able to

1. **Demonstrate** Historical Understanding.
2. **Analyze** Fundamental Rights and Duties
3. **Comprehend** Constitutional Institutions
4. **Evaluate** Constitutional Amendments and Contemporary Issues.

Detailed Syllabus:

Unit	Description	Duration (Hrs)
I	**Fundamentals of Indian Constitution** ● Introduction to the Indian Constitution: Historical background, making of the Constitution, and its significance ● Preamble: Understanding the importance and interpretation of the Preamble to the Constitution ● Fundamental Rights, Directive Principles of State Policy ● Analyze the salient features of the Indian Constitution	7
II	**Structure and Functioning of Government** ● Union Executive: Study of the President, Vice-President, Prime Minister, and Council of Ministers, along with their powers and functions ● Parliament: Understanding the composition, powers, and functioning of the Lok Sabha and Rajya Sabha ● Judiciary: Analysis of the structure, independence, and functioning of the Supreme Court and High Courts ● Federalism: Examination of the division of powers between the Union and States, along with the role of institutions like Governors and State Legislatures	7
III	**Constitutional Amendments and Legal Framework** ● Amendment Process: Understanding the procedure for amending the Constitution and significant amendments ● Constitutional Bodies: Study of institutions like the Election Commission, Comptroller and Auditor General, and their constitutional roles ● Emergency Provisions: Analysis of the provisions related to national emergency, state emergency, and financial emergency ● Constitutional Remedies: Detailed study of writs, judicial review, and other constitutional remedies available to citizens	8

Vision and Mission of the Computer Science & Engineering (AI&ML) Department

Vision:

To provide value-based technical Education in Computer Science and Engineering with specialization in Artificial Intelligence and Machine Learning.

Mission:

- To develop technically competent and innovative computer science engineers with in-depth knowledge of Artificial Intelligence and Machine Learning.
- To build ethically responsible, knowledgeable and skilled engineers to serve the needs of industry and society at large.
- To provide a conducive environment and opportunities for holistic development of students.